

UVA CS 4774: Machine Learning

Lecture 2: Machine Learning in a Nutshell

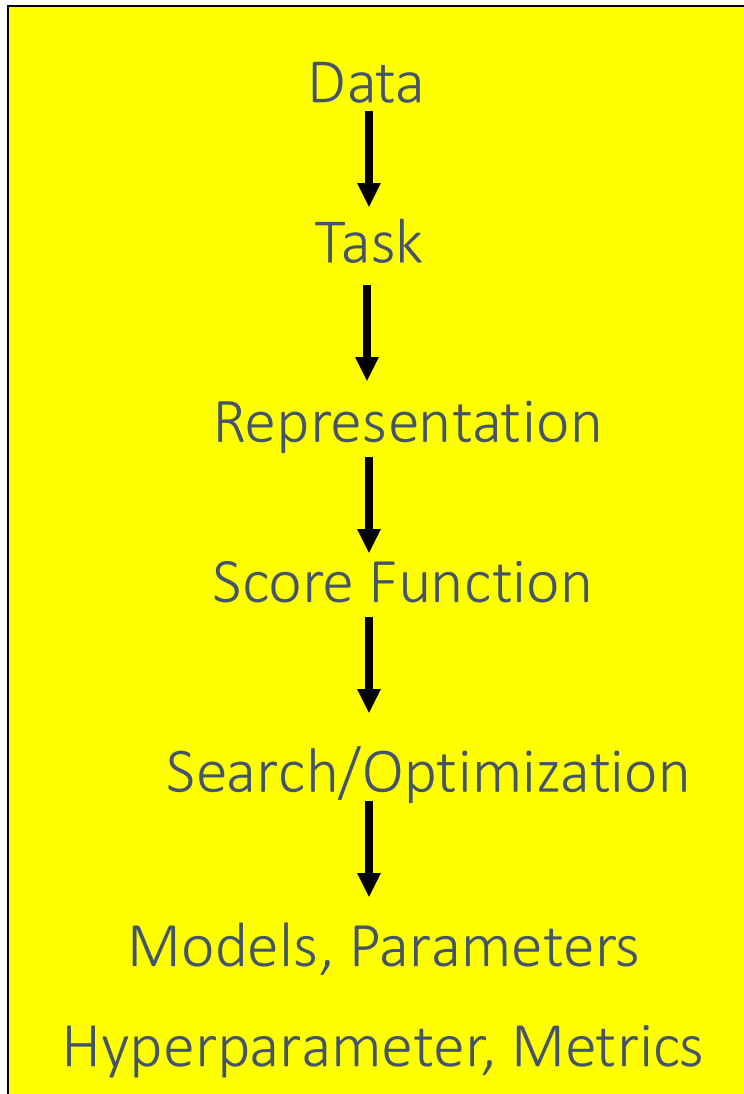
Dr. Yanjun Qi

University of Virginia
Department of Computer Science

Roadmap

- Machine Learning in a Nutshell
- Examples of Different Data Types
- Examples of Different Tasks
- Examples of Different Representation Types
- Examples of Different Loss/Cost Types
- Examples of Different Model Properties

Machine Learning in a Nutshell



ML grew out of
work in AI

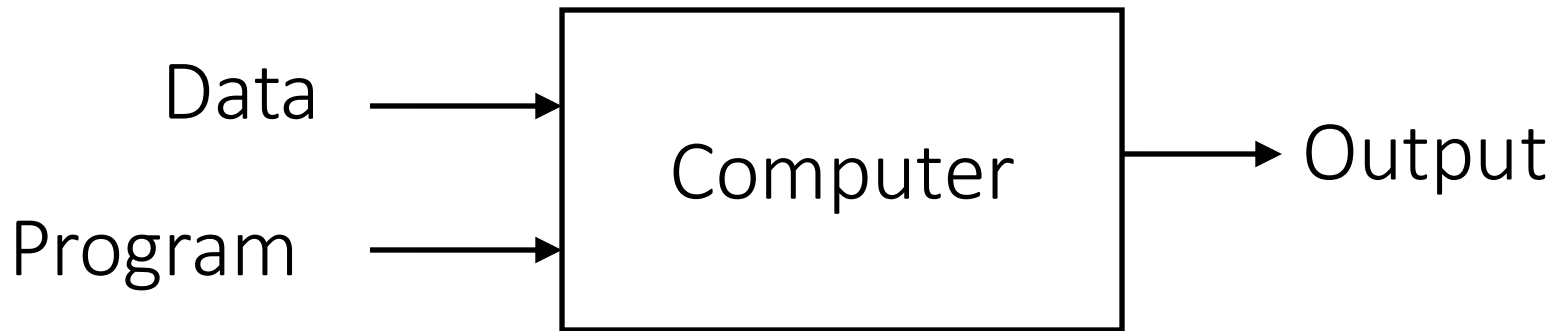
Optimize a
performance criterion
using example data or
past experience,

Aiming to generalize to
unseen data

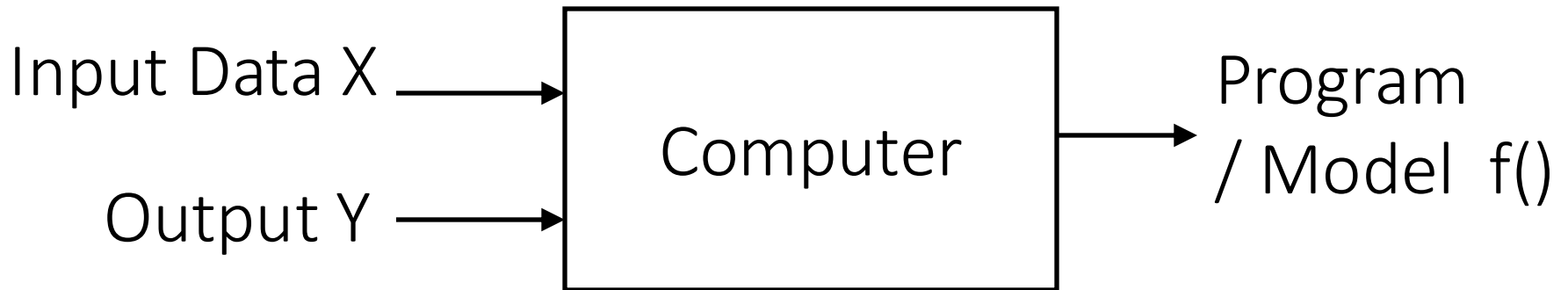
BASICS OF MACHINE LEARNING

- “The goal of machine learning is to build computer systems that can **learn and adapt from their experience.**” – Tom Dietterich
- “**Experience**” in the form of available **data examples** (also called as instances, samples)
- Available examples are described with properties (**data points in feature space X**)

Traditional Programming



Machine Learning (training phase)



Common Notations

- Inputs
 - p #input variables, n #observations
 - X : matrix written in bold capital
 - X_j (jth element of vector X)
 - i-th observed value of X_j is $X_{j,i}$ (a scalar)
 - Vectors are assumed to be column vectors
- Outputs
 - quantitative Y
 - qualitative C (for categorical)
- Observed variables / instances written in lower case

e.g. SUPERVISED LEARNING

- Find function to map **input** space X to **output** space Y

$$f : X \longrightarrow Y$$

- So that the **difference** between y and $f(x)$ of each example x is small.

e.g.

x	I believe that this book is not at all helpful since it does not explain thoroughly the material . it just provides the reader with tables and calculations that sometimes are not easily understood ...
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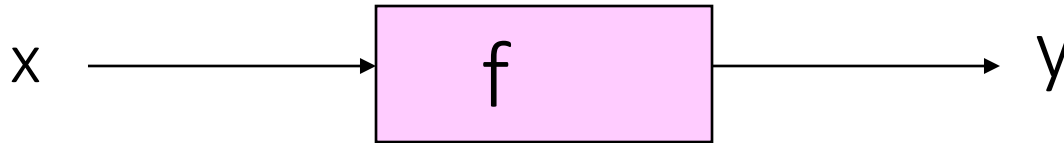
y	-1
---	----

Output Y: {1 / Yes , -1 / No }
e.g. Is this a positive product review ?

Input X : e.g. a piece of English text

e.g., SUPERVISED Linear Binary Classifier

- Now let us check out a VERY SIMPLE case of



e.g.: Binary y / Linear f / X as $\mathbb{R}^2$

$$f(x, w, b) = \text{sign}(w^T x + b)$$

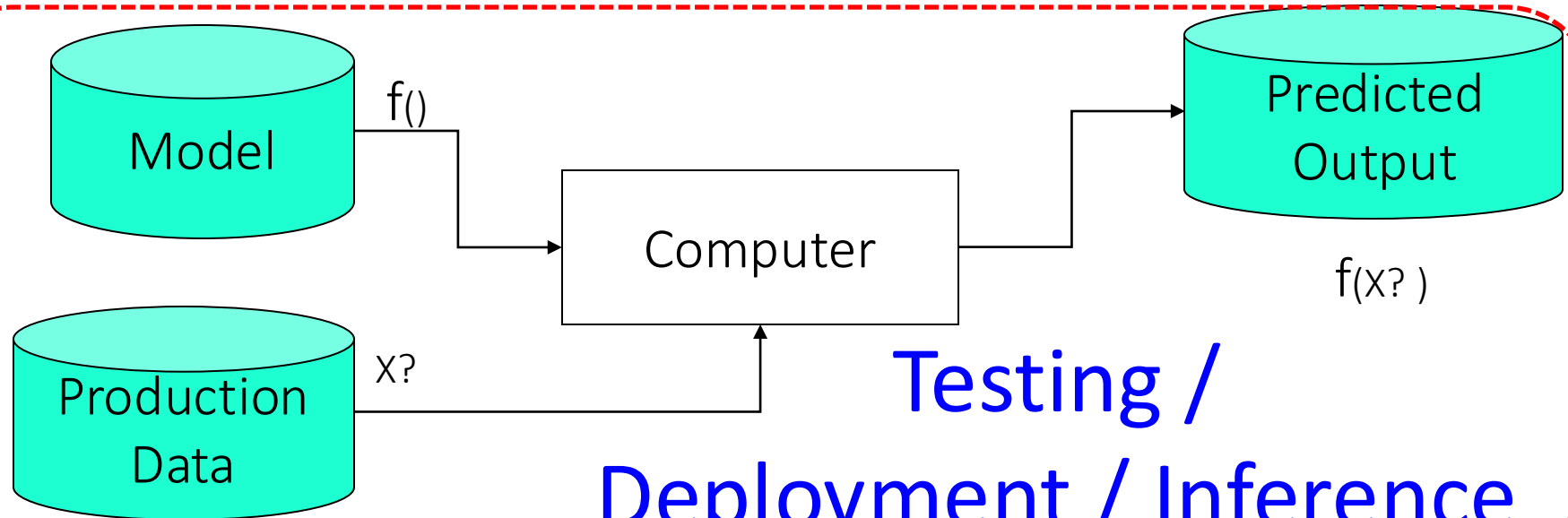
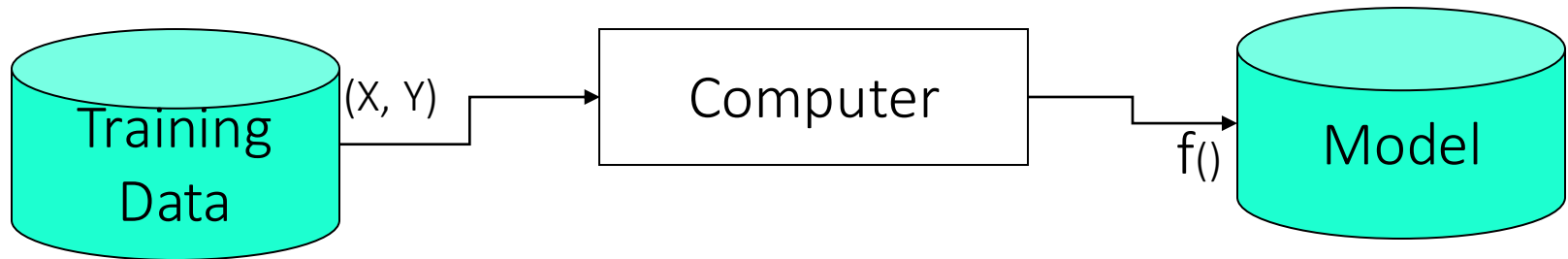
$$x^T = (x_1, x_2)$$

$$w^T = (w_1, w_2)$$

Two Phases (Modes) of Machine Learning

Consists of **input-output** pairs

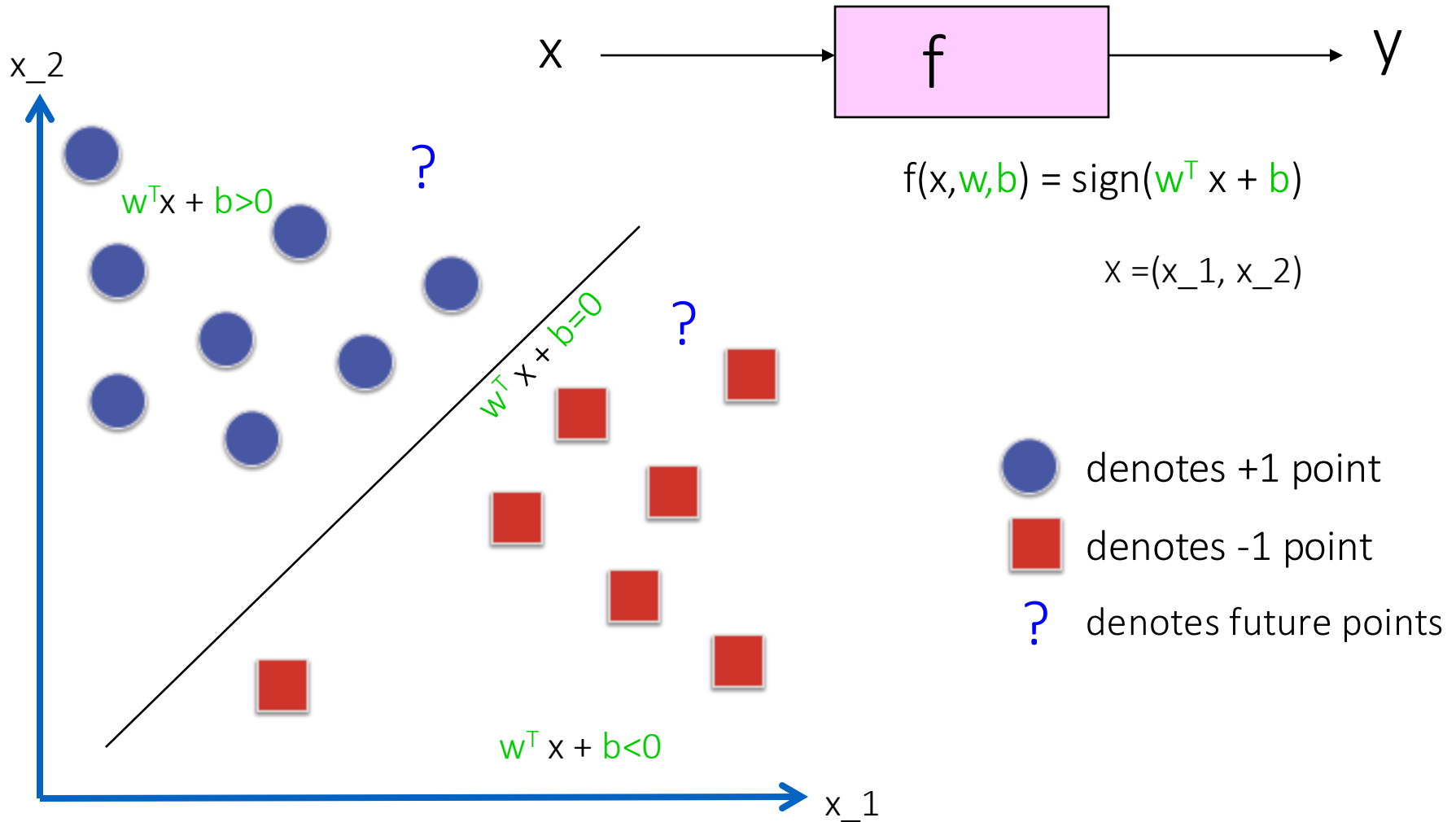
Training



Testing / Deployment / Inference

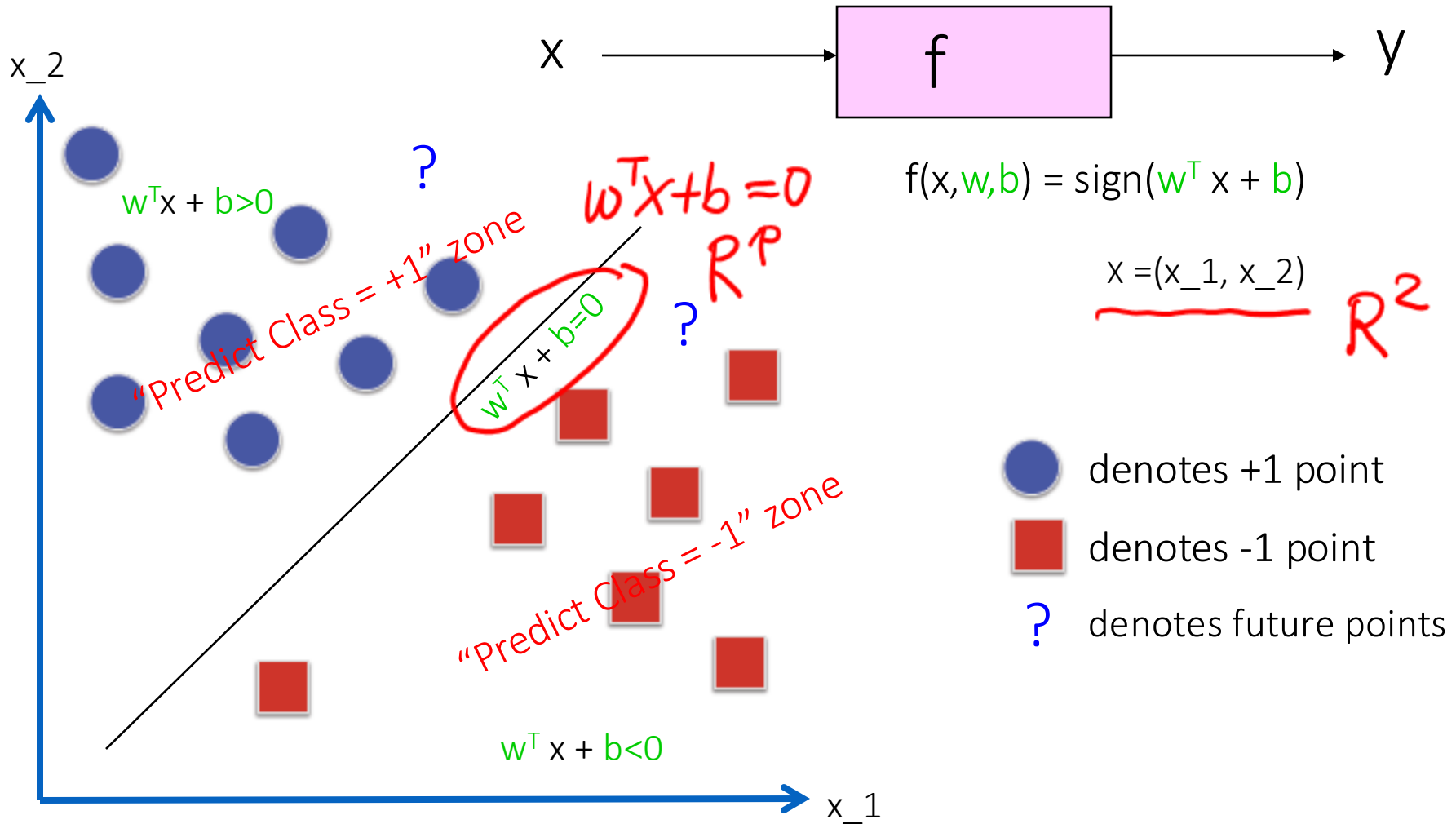
SUPERVISED Linear Binary Classifier

: Binary y / Linear f / X as $\mathbb{R}^2$



SUPERVISED Linear Binary Classifier

: Binary y / Linear f / X as $\mathbb{R}^2$



Basic Concepts

- Training (i.e. learning parameters w, b)
 - Training set includes
 - available examples $x_1, \dots, x_L$
 - available corresponding labels $y_1, \dots, y_L$
 - Find (w, b) by minimizing loss / Cost function $L()$
 - (i.e. difference between y and $f(x)$ on available examples in training set)

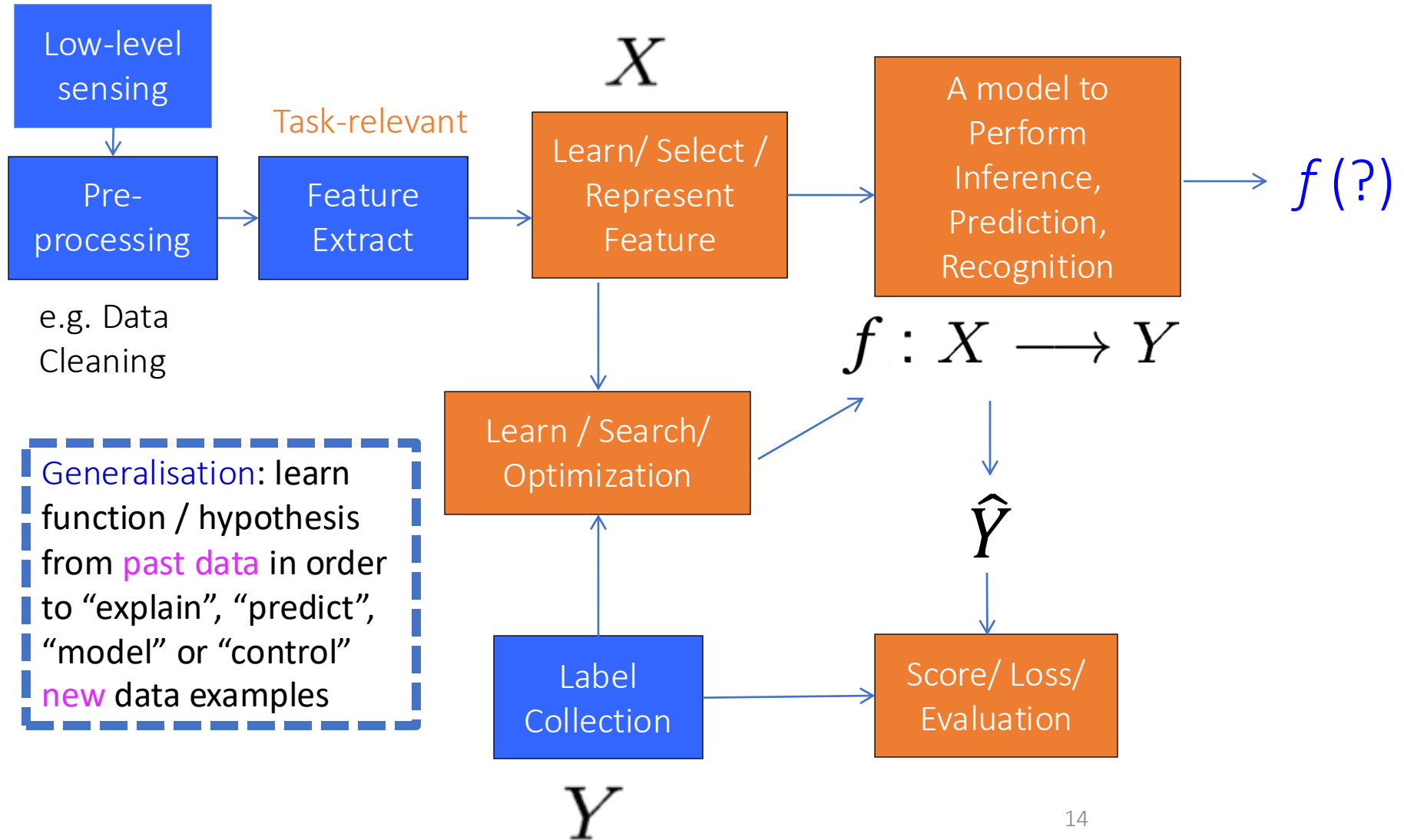
$$(W, b) = \underset{W, b}{\operatorname{argmin}} \sum_{i=1}^N \ell(\underbrace{f(x_i)}, \underbrace{y_i})$$

Handwritten red annotations: An arrow points from the symbol ℓ to the summation term. A bracket underlines the entire summation expression.

Basic Concepts

- **Testing** (i.e. evaluating performance on “future” points)
 - Difference between true $Y_?$ and the predicted $f(x_?)$ on a set of testing examples (i.e. testing set)
 - Key: example $x_?$ not in the training set
- **Generalisation**: learn function / hypothesis from past data in order to “explain”, “predict”, “model” or “control” new data examples

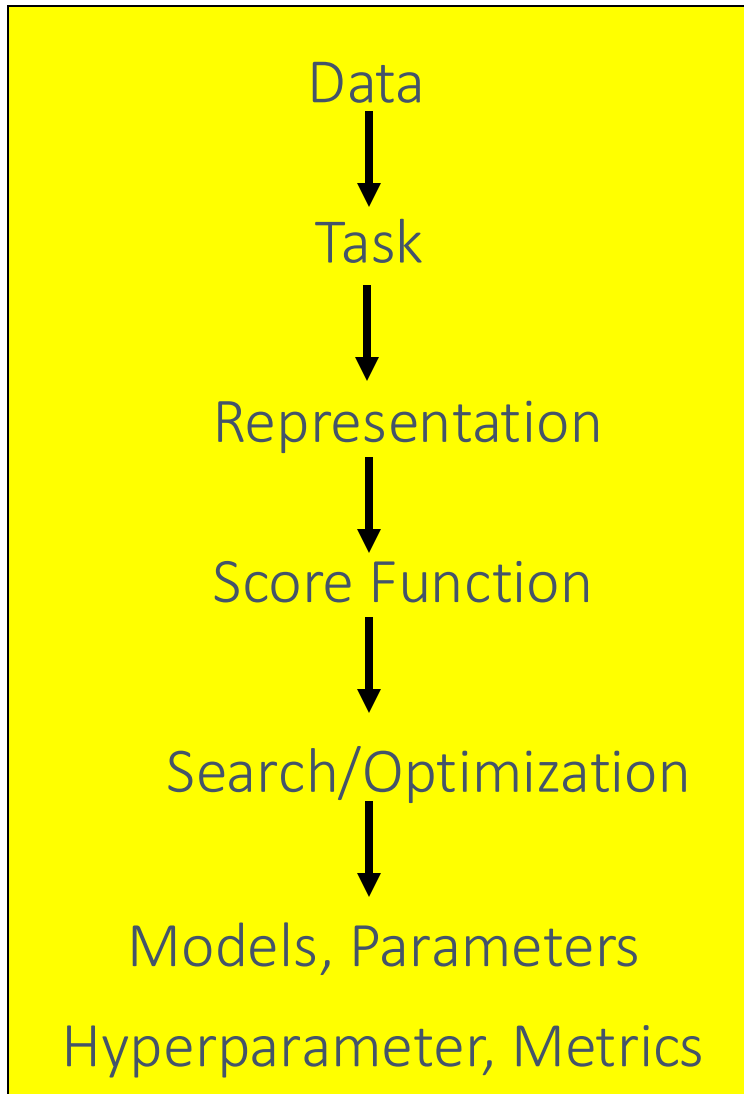
A Typical Machine Learning Application's Pipeline



When to use Machine Learning (Adapt to / learn from data) ?

- 1. Extract knowledge from data
 - Relationships and correlations can be hidden within large amounts of data
 - The amount of knowledge available about certain tasks is simply too large for explicit encoding (e.g. rules) by humans
- 2. Learn tasks that are difficult to formalise
 - Hard to be defined well, except by examples, e.g., face recognition
- 3. Create software that improves over time
 - New knowledge is constantly being discovered.
 - Rule or human encoding-based system is difficult to continuously re-design “by hand”.

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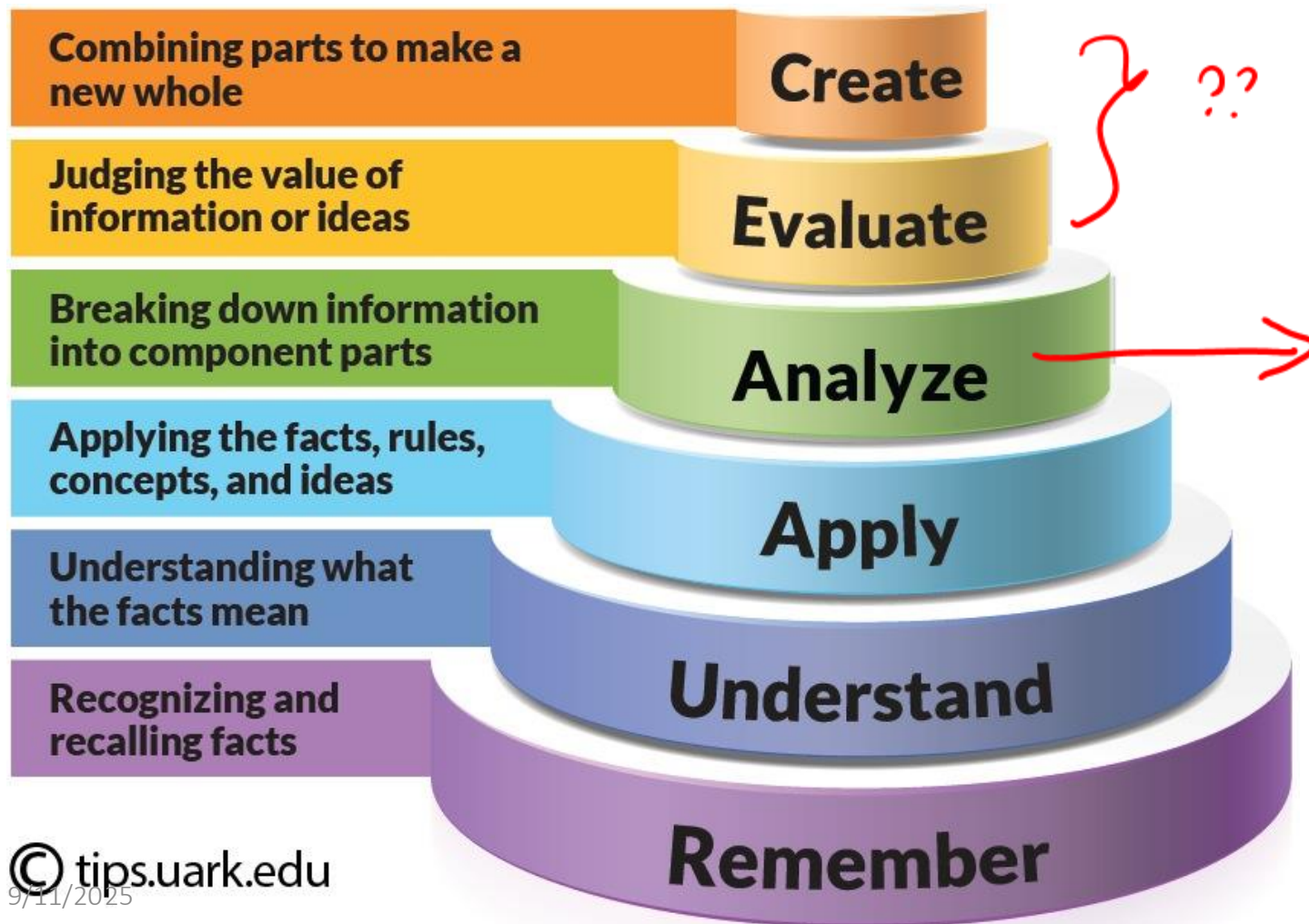
What we have covered

Data	
Task	
Representation	
Score Function	
Search/Optimization	
Models, Parameters	
9/11/2025	

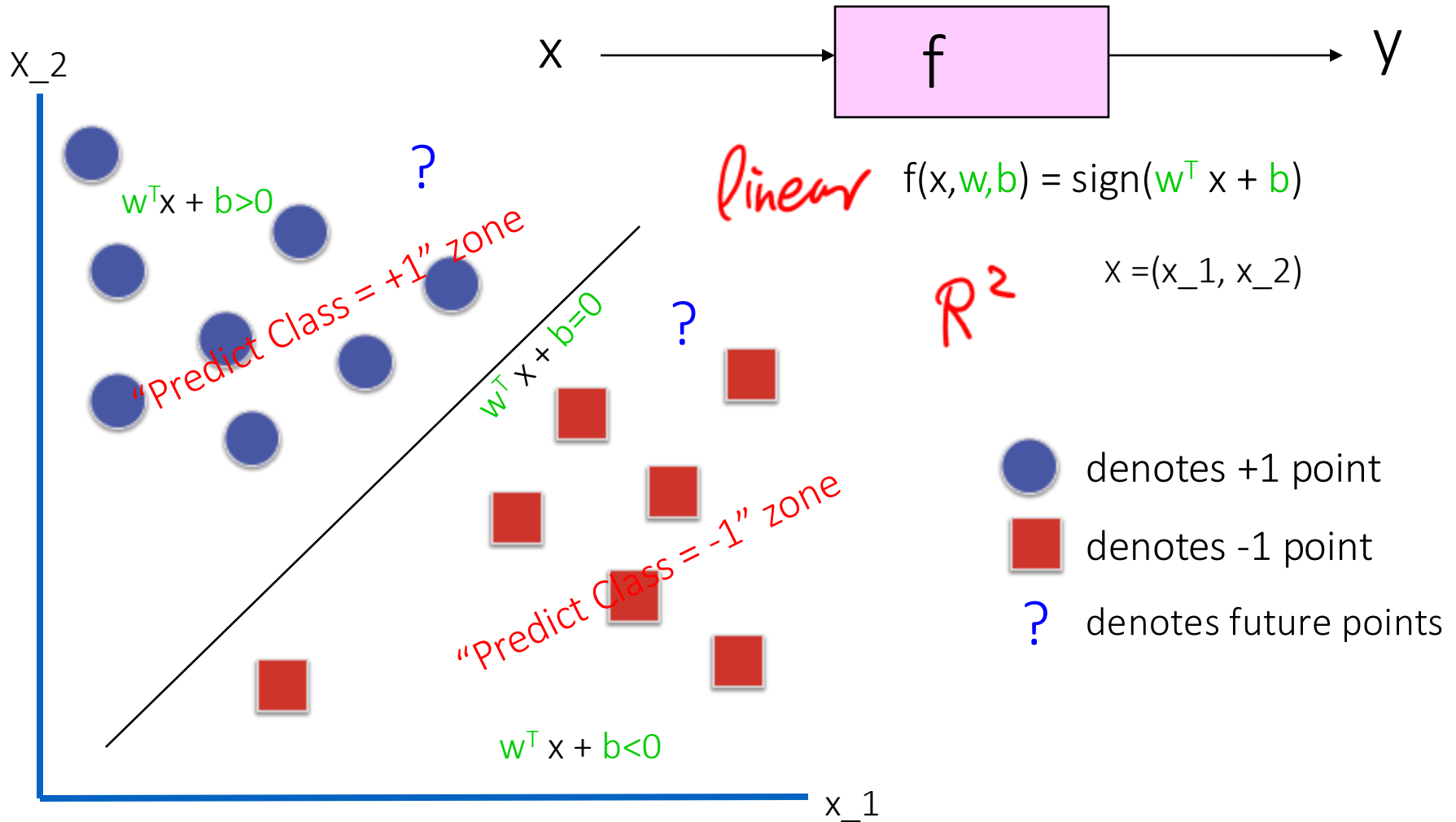
What we will cover

Data	Tabular, 1-D sequential, 2-D Grid like Imaging, 3-D VR, Graph, Set
Task	Regression, classification, clustering, dimen-reduction
Representation	Linear func, nonlinear function (e.g. polynomial expansion), local linear, logistic function (e.g. $p(c x)$), tree, multi-layer, prob-density family (e.g. Bernoulli, multinomial, Gaussian, mixture of Gaussians), local func smoothness, kernel matrix, local smoothness, partition of feature space,
Score Function	MSE, Margin, log-likelihood, EPE (e.g. L2 loss for KNN, 0-1 loss for Bayes classifier), cross-entropy, cluster points distance to centers, variance, conditional log-likelihood, complete data-likelihood, regularized loss func (e.g. L1, L2) , goodness of inter-cluster similar
Search/ Optimization	Normal equation, gradient descent, stochastic GD, Newton, Linear programming, Quadratic programming (quadratic objective with linear constraints), greedy, EM, asyn-SGD, eigenDecomp, backprop
Models, Parameters	Linear weight vector, basis weight vector, local weight vector, dual weights, training samples, tree-dendrogram, multi-layer weights, principle components, member (soft/hard) assignment, cluster centroid, cluster covariance (shape), ...

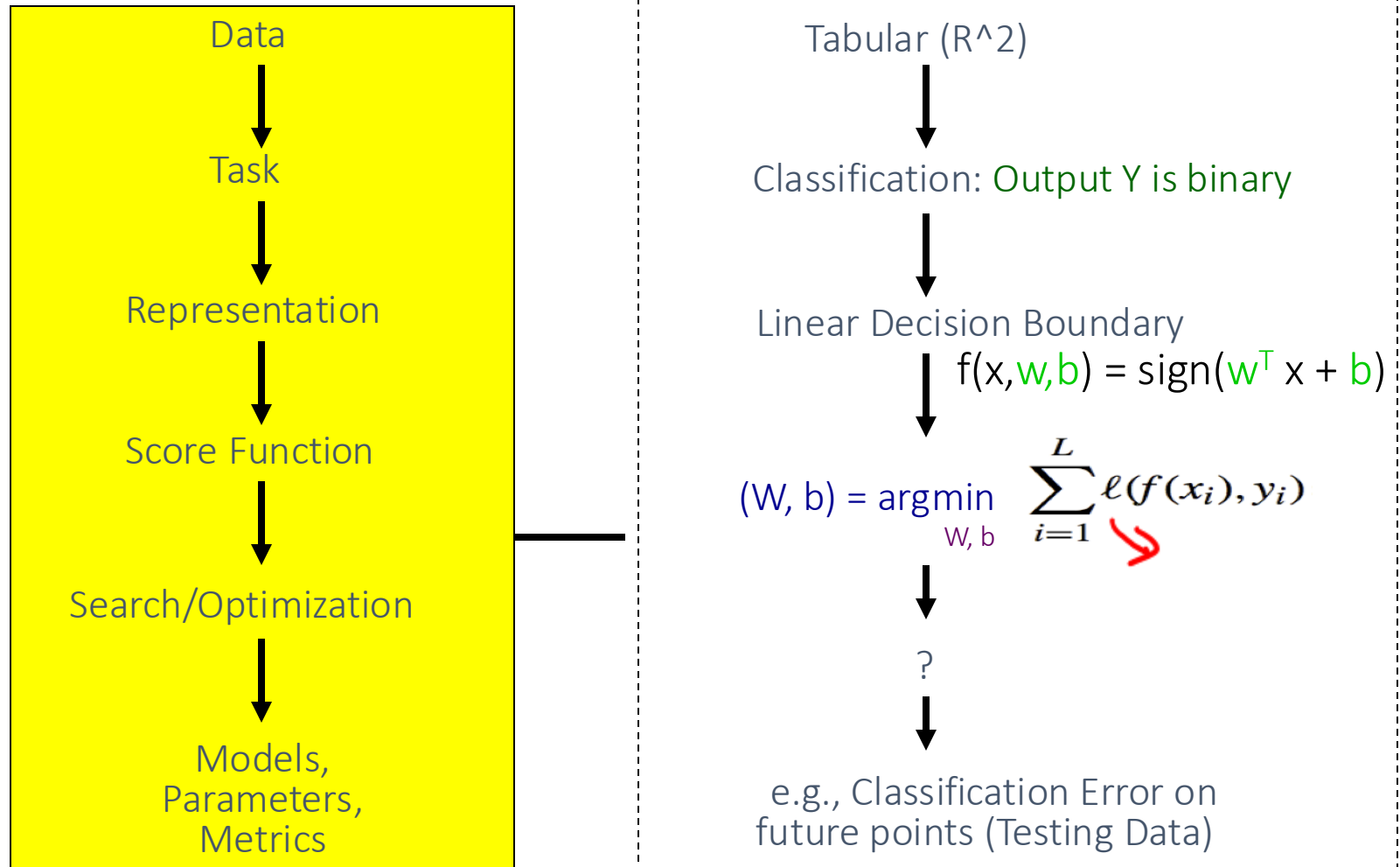
My Teaching Guide: Bloom's Taxonomy on Cognitive Learning



SUPERVISED Linear Binary Classifier



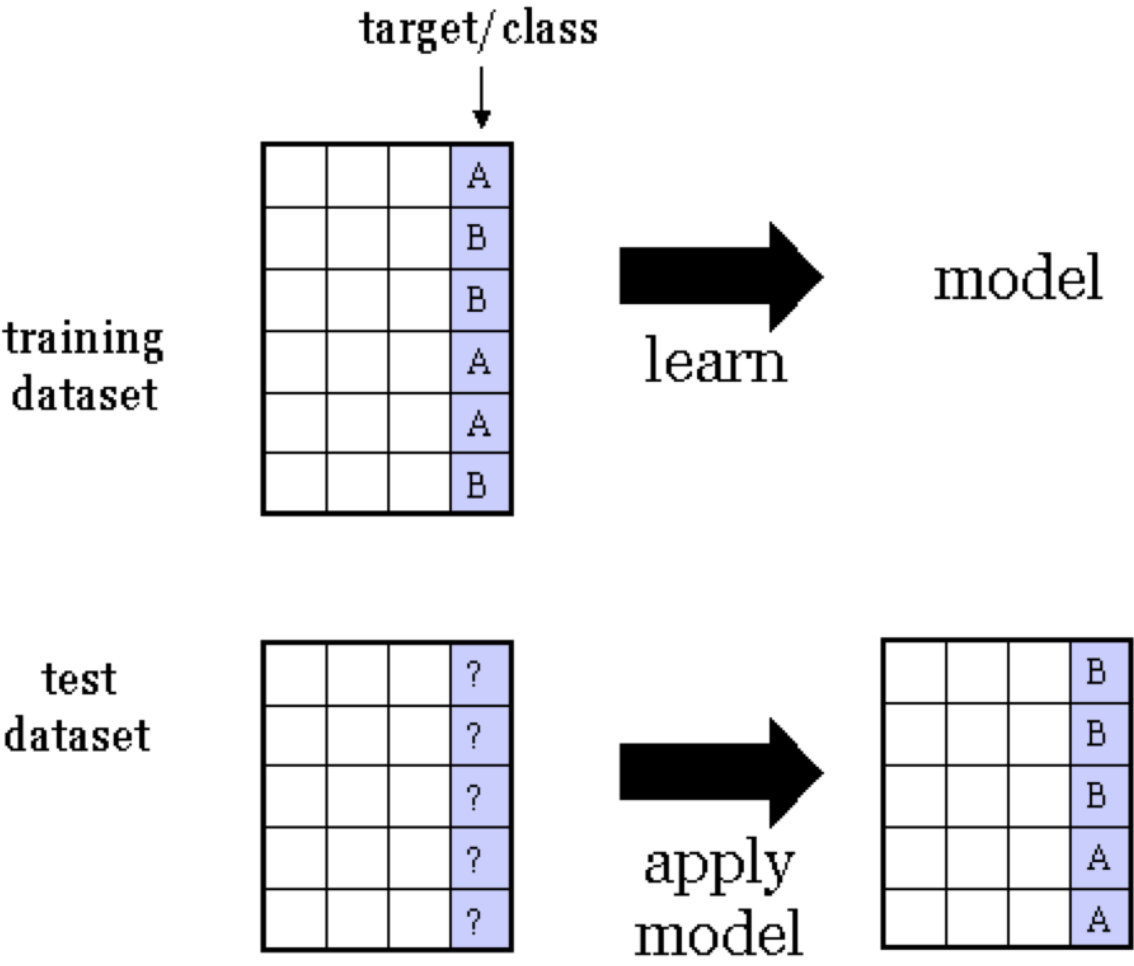
Nutshell for the simple Linear Supervised Classifier



<https://scikit-learn.org/stable/tutorial/basic/tutorial.html>

https://scikit-learn.org/stable/auto_examples/classification/plot_digits_classification.html#sphx-glr-auto-examples-classification-plot-digits-classification-py

I will run through notebook : Recognizing hand-written digits with L2.ipynb
Adapted from: ScikitLearn Tutorial plot_digits_classification.ipynb



Training dataset consists of **input-output** pairs

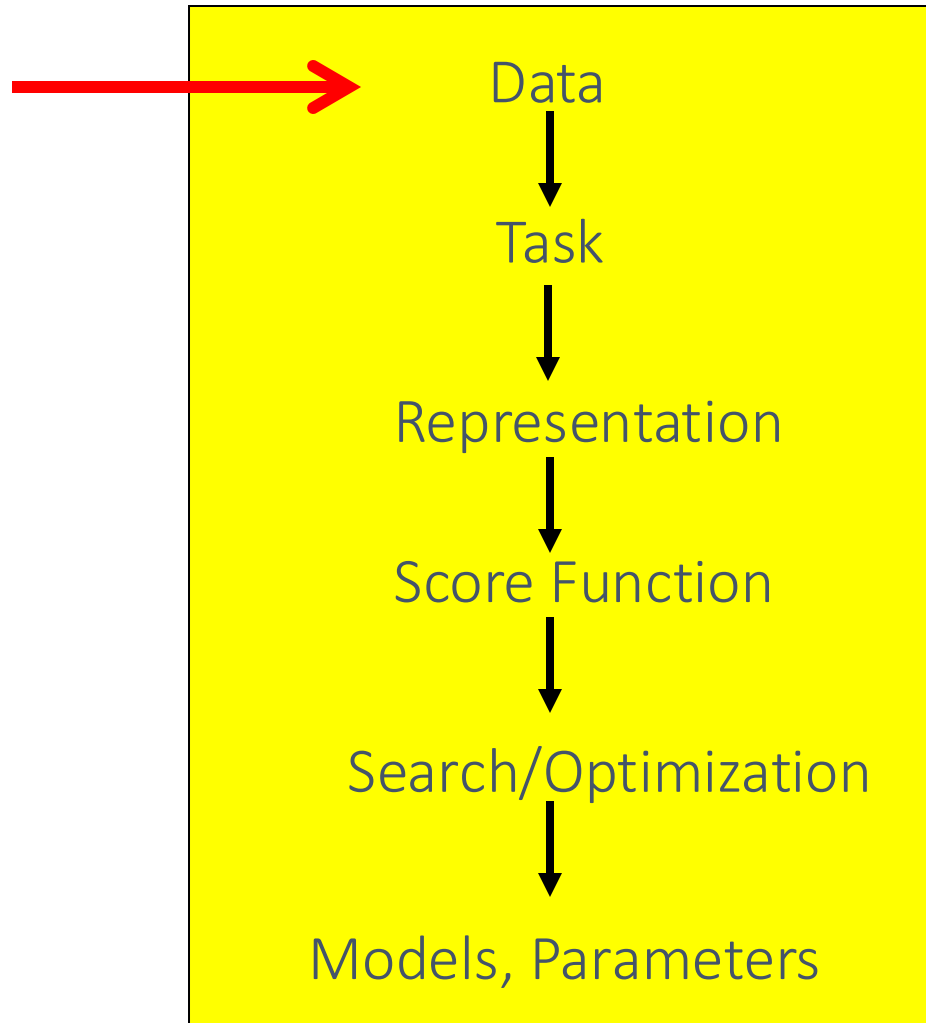
Thank You



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Optimize a
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	X_1	X_2	X_3	Y
s_1				
s_2				
s_3				
s_4				
s_5				
s_6				

t

$$f : \boxed{X} \longrightarrow \boxed{Y}$$

- **Data**/points/instances/examples/samples/records: [rows]
- **Features**/attributes/dimensions/independent variables/covariates/predictors/regressors: [columns, except the last]
- **Target**/outcome/response/label/dependent variable: special column to be predicted [last column]

	X_1	X_2	X_3	Y
s_1				
s_2				
s_3				
s_4				
s_5				
s_6				

t

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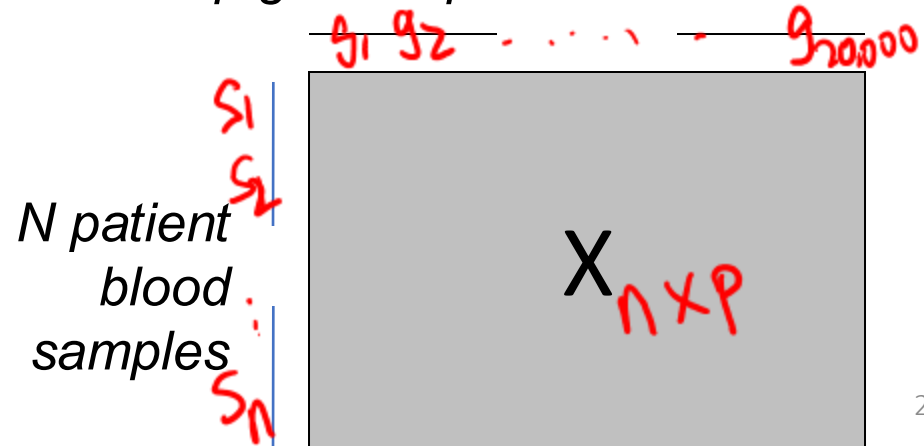
Main Types of Columns

	X_1	X_2	X_3	Y
S_1				
S_2				
S_3				
S_4				
S_5				
S_6				

- *Continuous*: a real number, for example, weight

- *Discrete*: a symbol, like “Good” or “Bad”

p genes' quantities in blood cell



training dataset 

$$\mathbf{X}_{train} = \begin{bmatrix} - & - & \mathbf{x}_1^T & - & - \\ - & - & \mathbf{x}_2^T & - & - \\ \vdots & & \vdots & & \vdots \\ - & - & \mathbf{x}_n^T & - & - \end{bmatrix}$$

$$\bar{\mathbf{y}}_{train} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

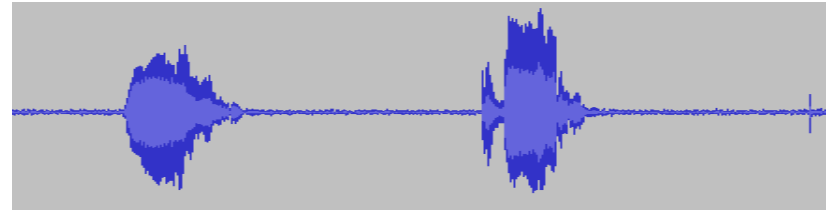
test dataset 

$$\mathbf{X}_{test} = \begin{bmatrix} - & - & \mathbf{x}_{n+1}^T & - & - \\ - & - & \mathbf{x}_{n+2}^T & - & - \\ \vdots & & \vdots & & \vdots \\ - & - & \mathbf{x}_{n+m}^T & - & - \end{bmatrix}$$

$$\bar{\mathbf{y}}_{test} = \begin{bmatrix} y_{n+1} \\ y_{n+2} \\ \vdots \\ y_{n+m} \end{bmatrix}$$

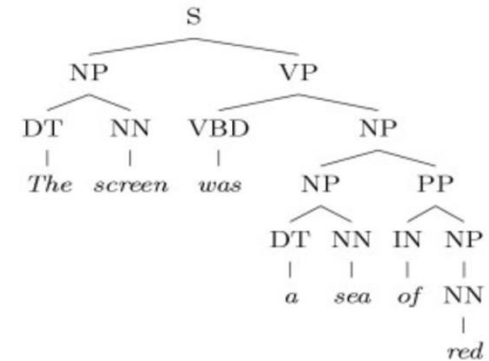
1D Sequence Data Type (eg. Language, Genome, Audio)

X I believe that this book is not at all helpful since it does not explain thoroughly the material . it just provides the reader with tables and calculations that sometimes are not easily understood ...

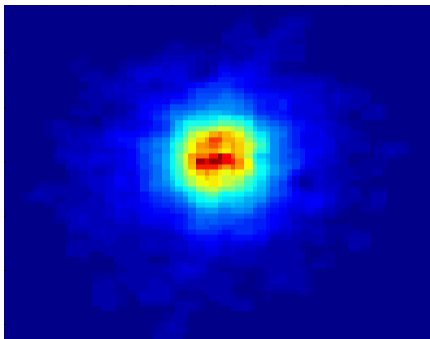


The screen was
a sea of red

Language Parsing



2D Grid Data Type (eg. Images)



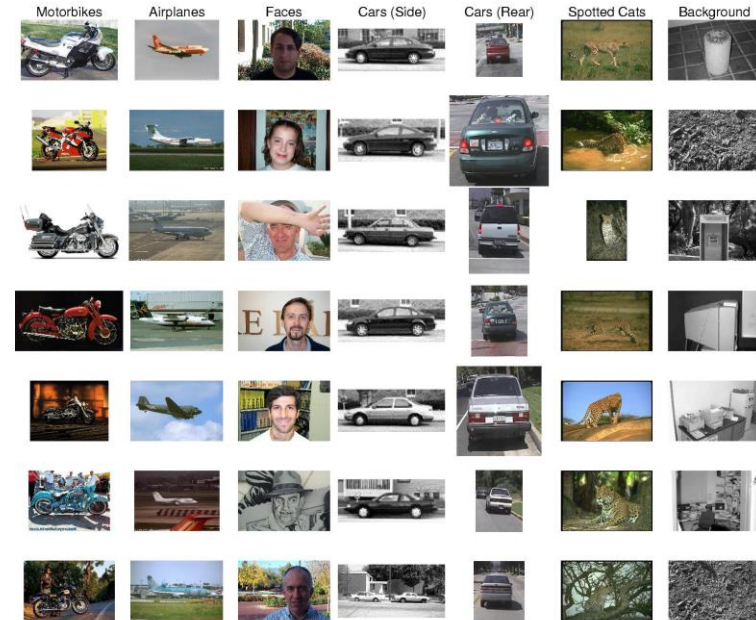
e.g.,

- 72 million stars, 20 million galaxies
- Object Catalog: 9 GB
- Image Database: 150 GB



Figure S6 Illustrative Examples of Chest X-Rays in Patients with Pneumonia,

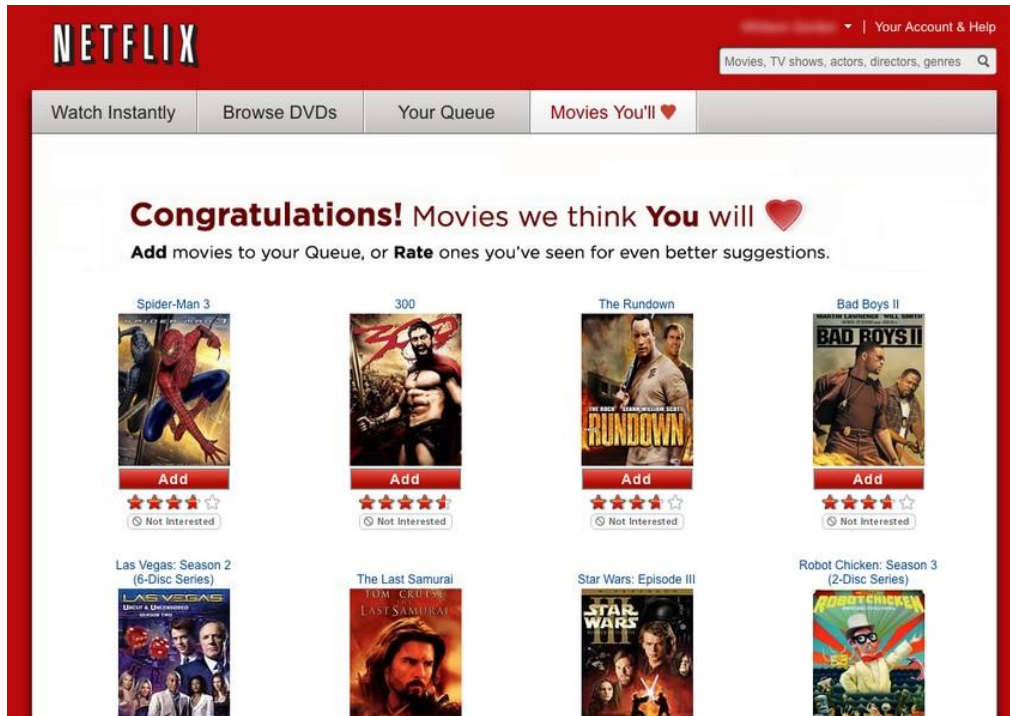
Kaggle: 5,232 chest X-ray images from children, including 3,883 characterized as depicting pneumonia (2,538 bacterial and 1,345 viral) and 1,349 normal, from a total of 5,856 patients



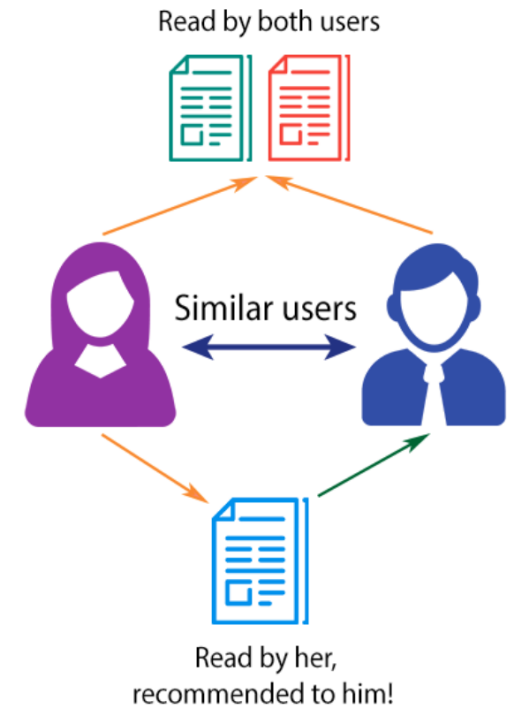
ImageNet Competition:

[Training on 1.2 million images [X]
vs. 1000 different word labels [Y]]

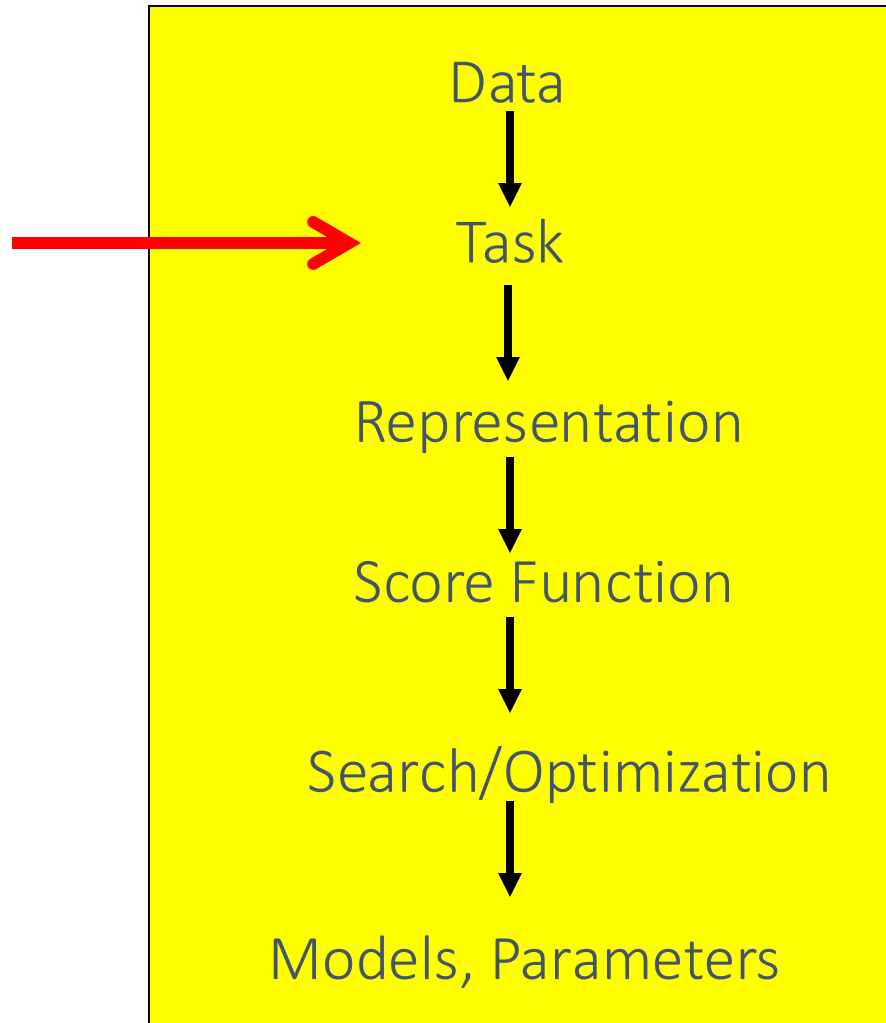
Graph Data Type (eg. Social Network)



COLLABORATIVE FILTERING



Machine Learning in a Nutshell

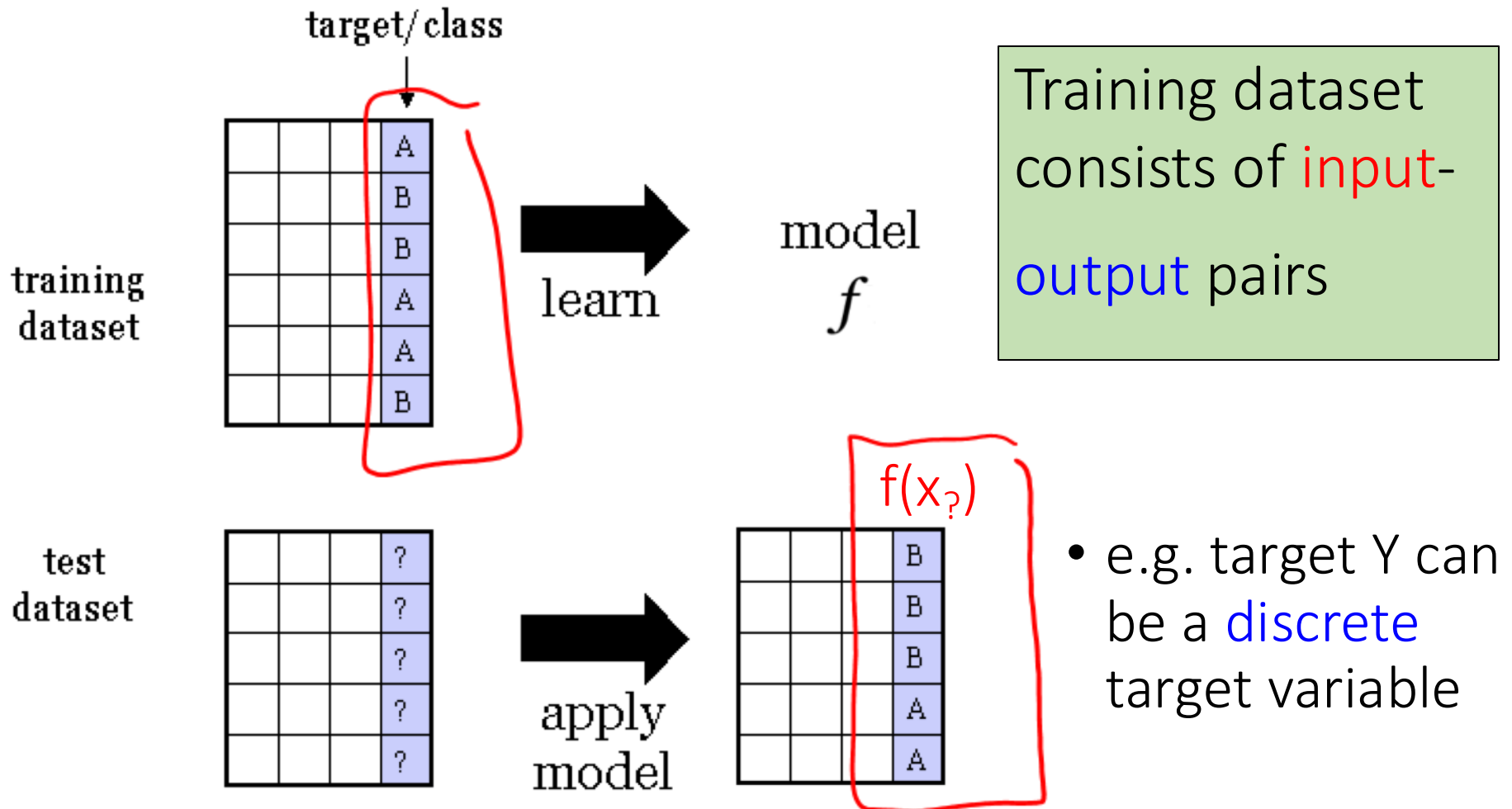


ML grew out of
work in AI

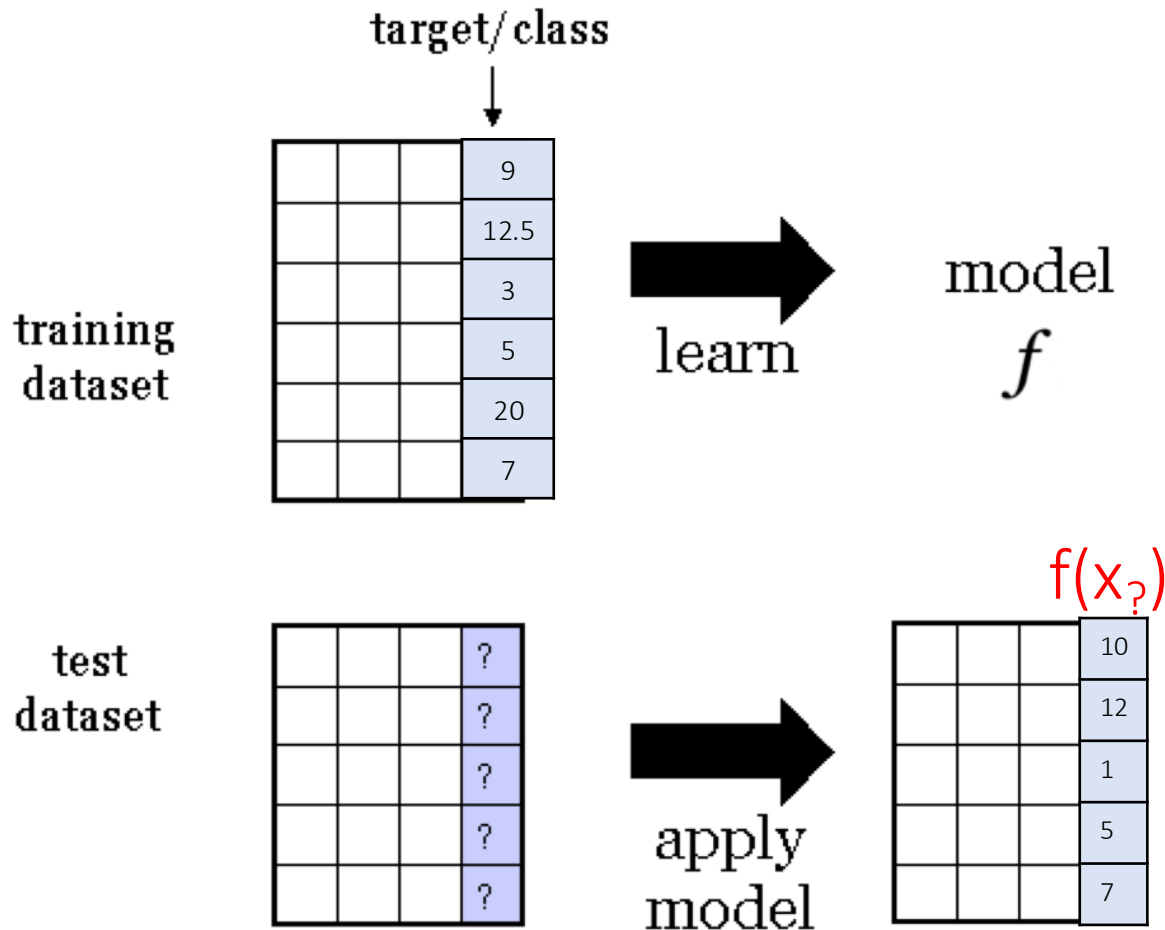
Optimize a
performance criterion
using example data or
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Aiming to generalize to
unseen data

e.g. SUPERVISED Classification



e.g. SUPERVISED Regression

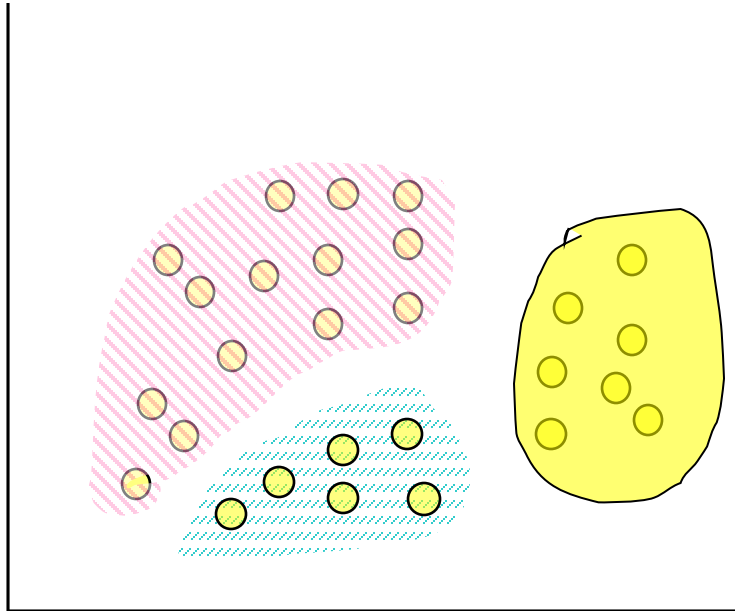


Training dataset consists of **input-output** pairs

- e.g. target Y can be a **continuous** target variable

Unsupervised LEARNING : [No Given Y]

- No labels are provided (e.g. No Y provided)
- Find patterns from unlabeled data, e.g. clustering

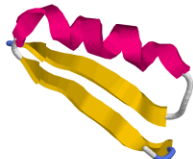
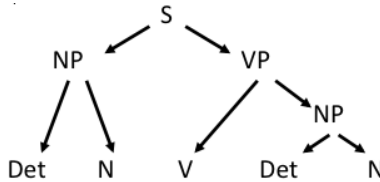


e.g. clustering => to find “natural” grouping of instances given unlabeled data

Structured Output LEARNING : [Complex Y]

- Many prediction tasks involve **output labels having structured correlations or constraints among instances**

Structured Dependency
between Examples' Y

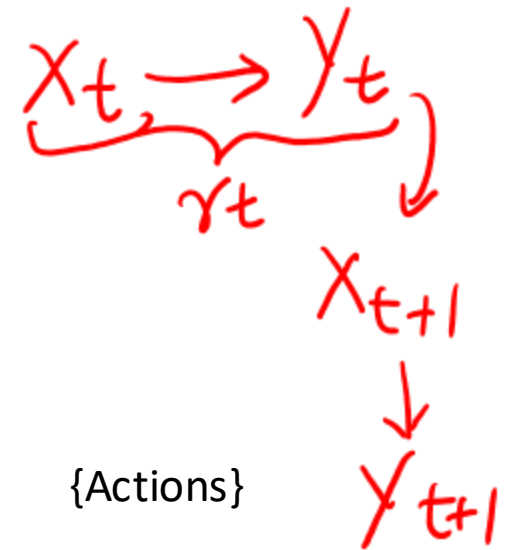
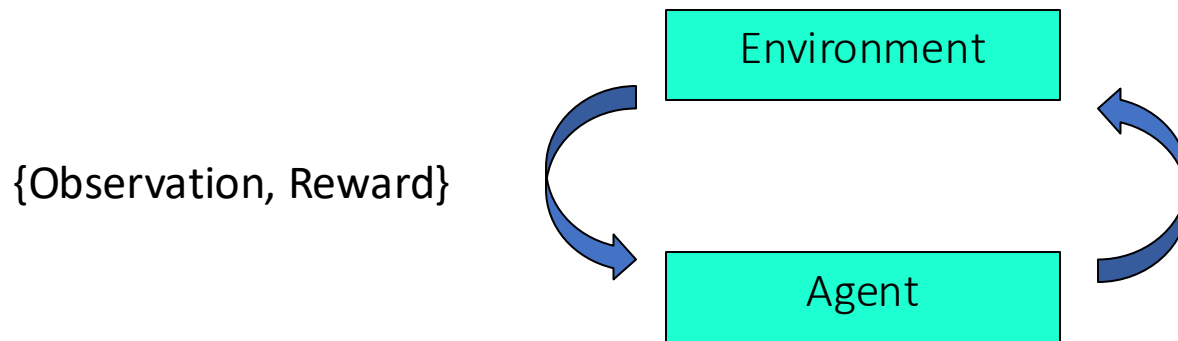
	Sequence	Tree	Grid
Input X	APAFSVSPASGACGPECA...	The dog chased the cat	
Output Y	 CCEEEECCCCHHHCCC...		

Many more possible structures between y_i , e.g. **spatial**, **temporal**, **relational** ...

Reinforcement Learning (RL)

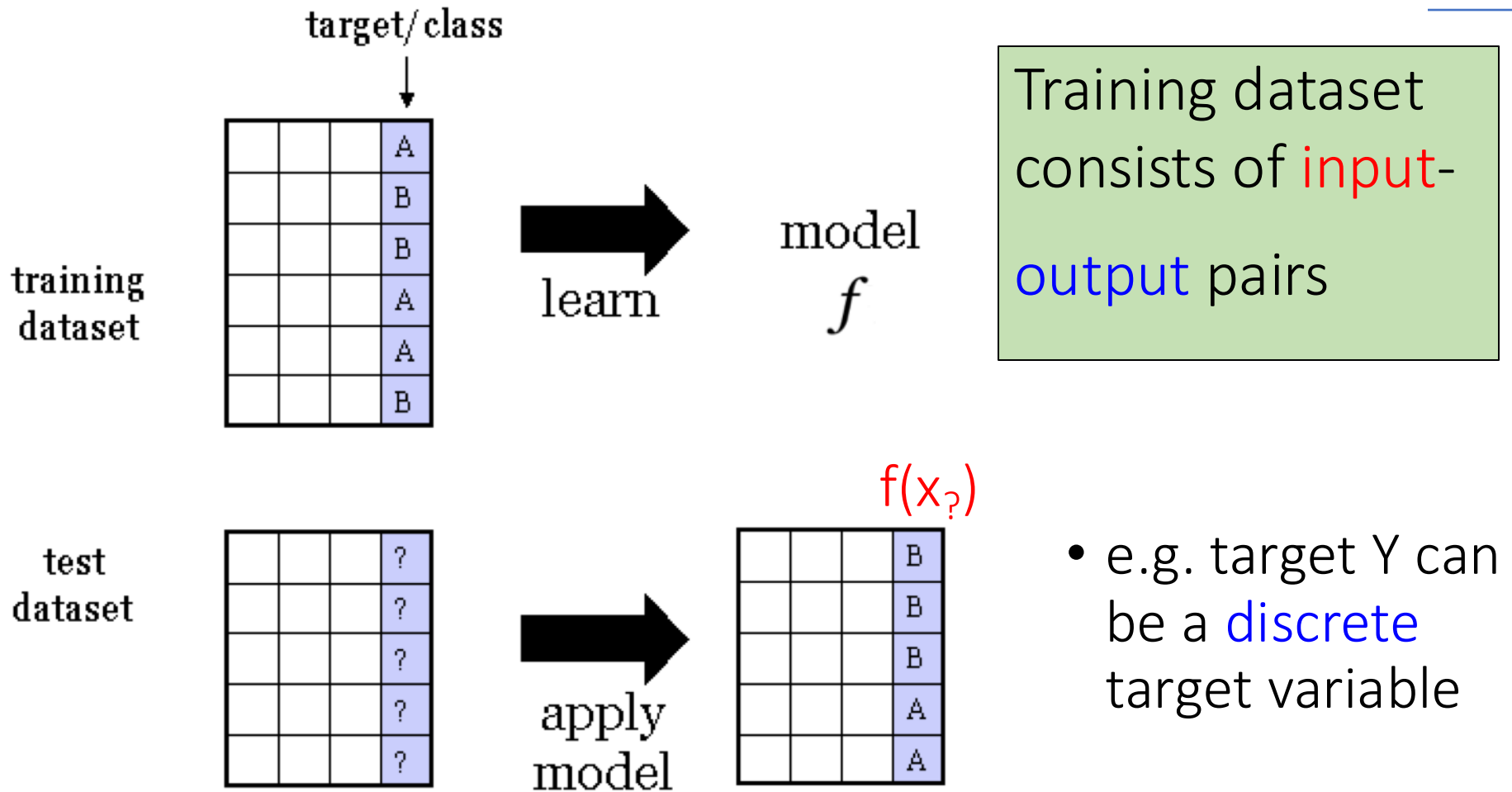
[Not IID, Sequential]

- What's Reinforcement Learning?



- Agent interacts with an environment and learns by maximizing a scalar reward signal
 - Basic version: No labels or any other supervision signal.
 - Variation like imitation learning: supervised

(Most basic:) SUPERVISED Classification



Many Variants of SUPERVISED Classification

- Binary Classification
- Multi-class Classification
- Hierarchical Classification
- Multi-label Classification
- Structured Predictions
-

Binary: Text Review-based Sentiment Classification

x	I believe that this book is not at all helpful since it does not explain thoroughly the material . it just provides the reader with tables and calculations that sometimes are not easily understood ...
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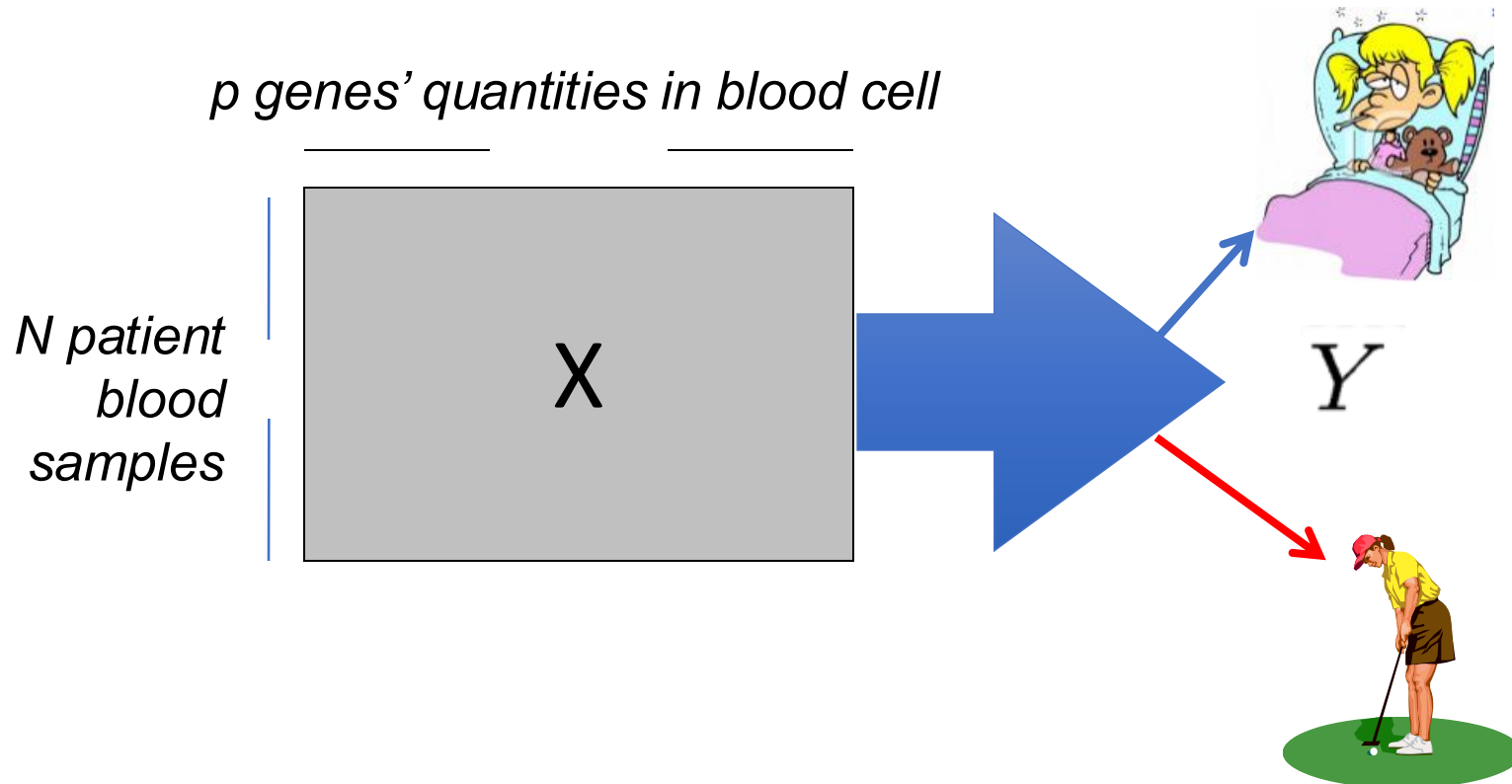
Input X : e.g. a piece of English text

y	-1
---	----

Output Y: {1 / Yes , -1 / No }

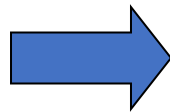
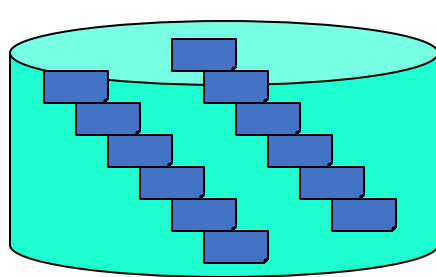
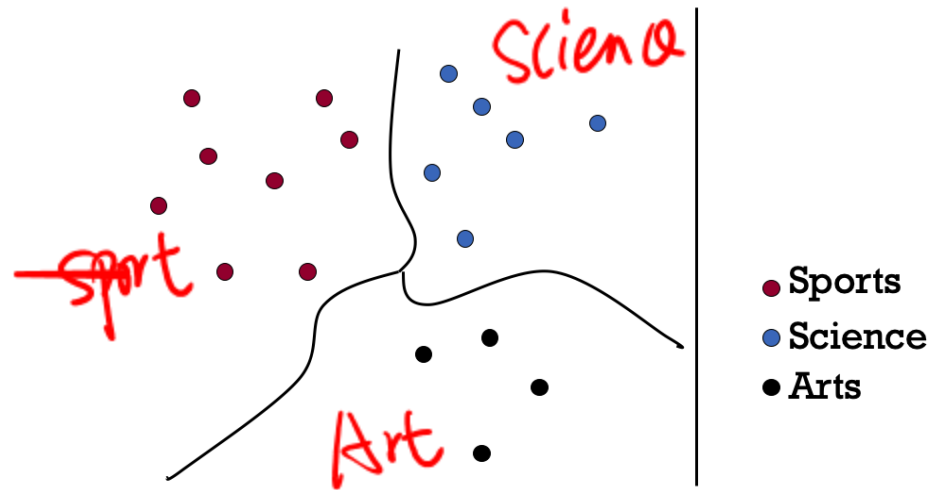
e.g. Is this a positive product review ?

Binary: : Disease Classification using gene expression

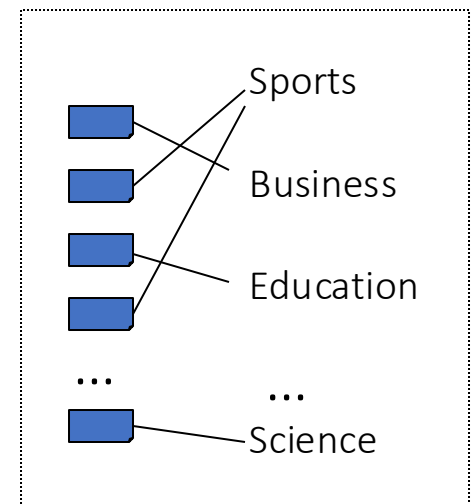
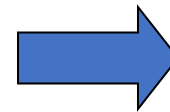


Multi-Class: Text Categorization

- Almost the most basic/standard supervised classification problem



Text
Categorization
System



$$\hat{y} = f(x)$$

Multi Label Classification (MLC)

- MLC is the task of assigning a set of target labels for a given sample
- Given input $\mathbf{x}$, predict the set of labels $\{y_1, y_2, \dots, y_L\}$, $y_i \in \{0, 1\}$

$\mathbf{x}$



y_1	Castle	✓
y_2	City	✗
y_3	Mountains	✓
y_4	Car	✗
y_5	Road	✓

Hierarchical: Text Categorization, e.g. Google News

The screenshot displays the Google News interface. On the left, a sidebar lists various news categories. A red bracket groups the following categories: World, U.S., Business, Technology, iPhone, Microsoft Windows, Minecraft, Safety, IBM, General Motors, Facebook, Microsoft Corporation, Tablet computers, and Tor. Below these are Entertainment, Sports, Science, Health, and Spotlight. The main content area features a 'Technology' section with a large article titled 'Microsoft Keyboard Works With Windows, iOS, and Android'. This article includes a photo of the keyboard, a 'See realtime coverage' button, and a list of related links. Below the main article, there are five smaller image thumbnails with captions: BetaNews, PhoneDog, SlashGear, WinBeta, and Hot Hardware. Further down, two more news items are visible: 'Microsoft/Minecraft Deal Gets a Skit On Conan O'Brien's Show' and 'Apple's iOS 8 available Wednesday'.

Google

Search News Search the Web

Search and browse 4,500 news sources updated continuously.

News

Top Stories
News near you

World
U.S.
Business
Technology
iPhone
Microsoft Windows
Minecraft
Safety
IBM
General Motors
Facebook
Microsoft Corporation
Tablet computers
Tor
Entertainment
Sports
Science
Health
Spotlight

Technology

Microsoft Keyboard Works With Windows, iOS, and Android

PC Magazine - 53 minutes ago

With a handful of new peripherals, Microsoft is revamping older products and embracing the new mobile reality. Oshares. Microsoft Universal Mobile Keyboard.

Microsoft announces new line of accessories for Windows, Android, iOS, and ... BetaNews

Microsoft's new Universal Mobile Keyboard works with iOS, Android and ... ZDNet

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See realtime coverage

BetaNews PhoneDog SlashGear WinBeta Hot Hardware

Microsoft/Minecraft Deal Gets a Skit On Conan O'Brien's Show

GameSpot - 1 hour ago

During Monday's episode of Conan, the comedian aired a segment about how the inventor of Minecraft would be celebrating the massive pay day.

Apple's iOS 8 available Wednesday

New York Daily News - 15 minutes ago

You don't need to order an iPhone 6 to feel like you've gotten a brand new phone. Apple's much-anticipated operating system update, iOS 8, will be available for download Wednesday.

IBM Watson Data Analysis Service Revealed

Generating: Text2Image

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



the flower has petals that are bright pinkish purple with white stigma



this white and yellow flower have thin white petals and a round yellow stamen



Open AI : DALL·E: Creating Images from Text

TEXT PROMPT

an illustration of a baby daikon radish in a tutu walking a dog

AI-GENERATED IMAGES



[Edit prompt or view more images](#) ↓

TEXT PROMPT

an armchair in the shape of an avocado. . . .

AI-GENERATED IMAGES



Generating : Edges2Image



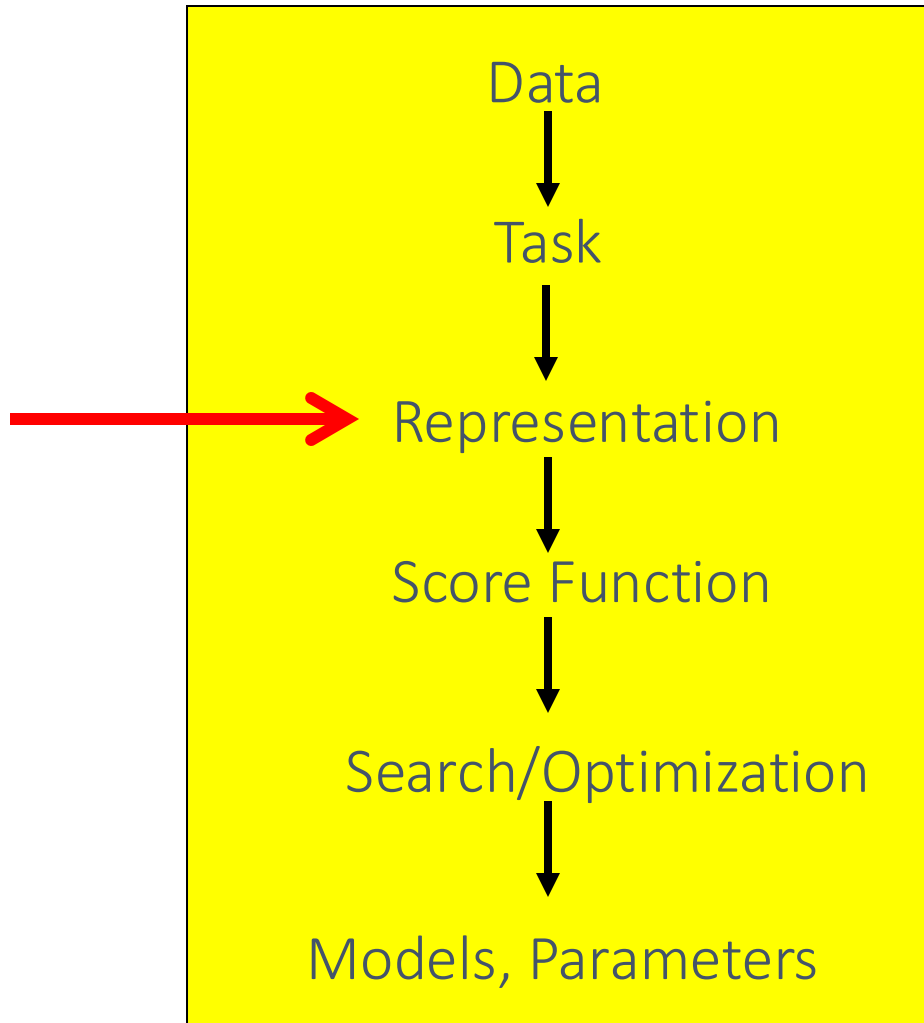
Thank You



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unseen data

- Text / String / Symbolic
- Sequences / Sets / Graph
 - Variable length
 - Discrete
 - Combinatorial
 - Spatial ordering among units



	X_1	X_2	X_3
s_1			
s_2			
s_3			
s_4			
s_5			
s_6			

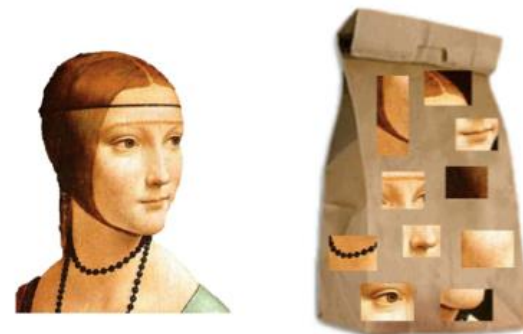
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Vector Space Representation: Bag of Words Trick

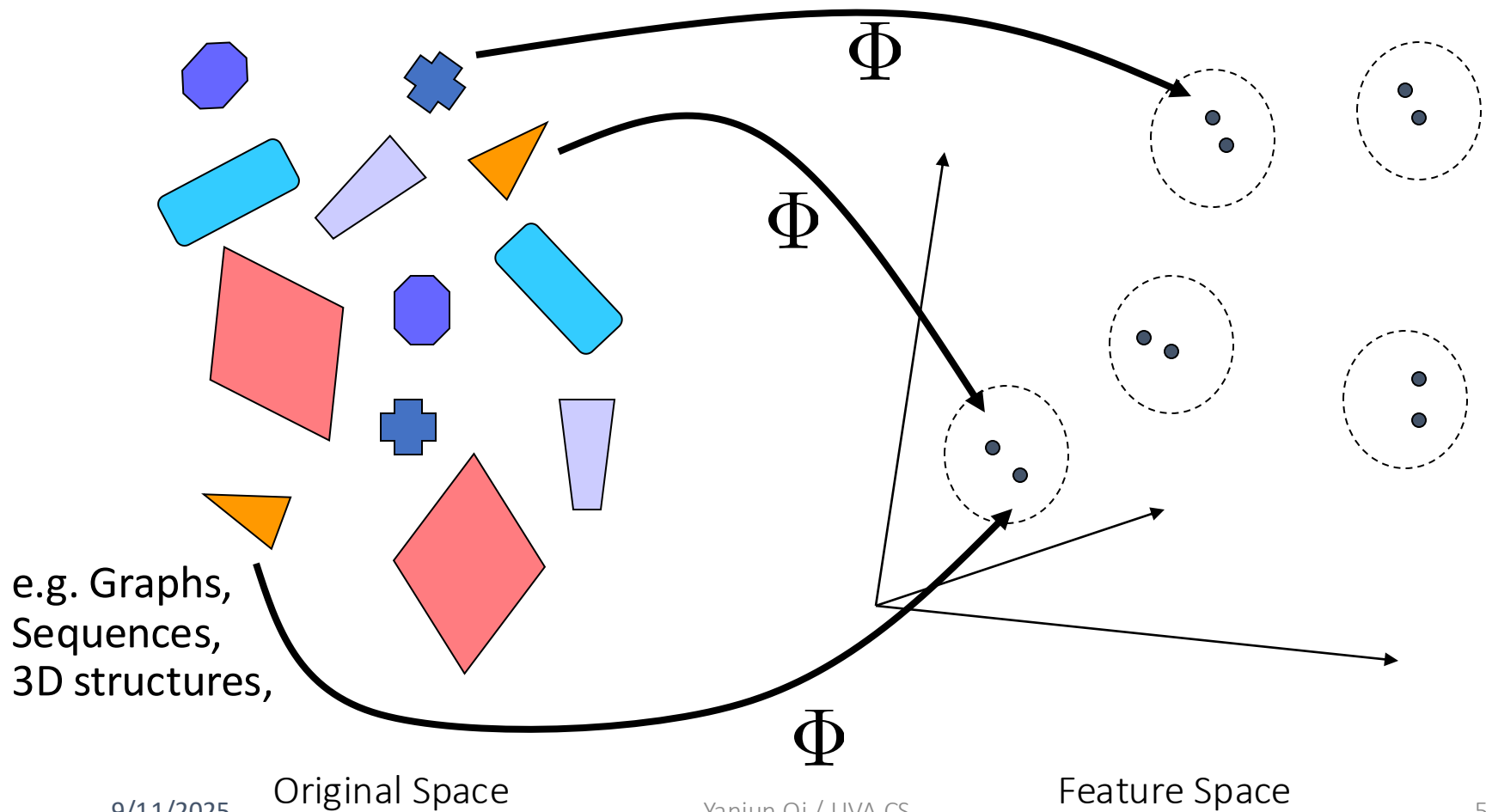
- Each document is a vector, one component for each term (= word).

	Doc 1	Doc 2	Doc 3	...
Word 1	3	0	0	...
Word 2	0	8	1	...
Word 3	12	1	10	...
...	0	1	3	...
...	0	0	0	...

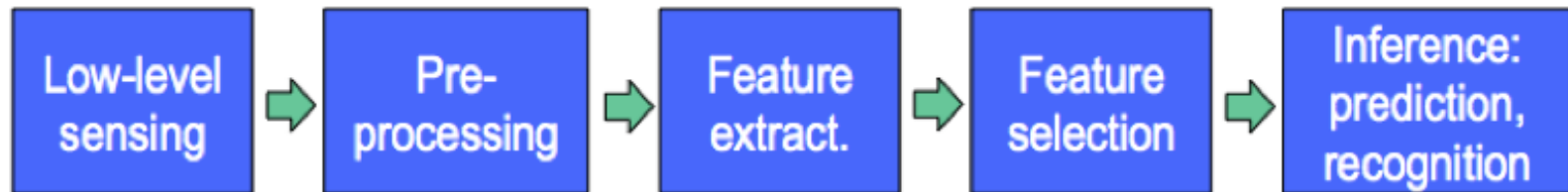
- Normalize to unit length.
- High-dimensional vector space:
 - Terms are axes, 10,000+ dimensions, or even 100,000+
 - Docs are vectors in this space



[Complex X] : Representation Learning!!!



DEEP LEARNING / FEATURE LEARNING :



Feature Engineering (before 2012)

- ✓ Most critical for accuracy
- ✓ Account for **most of the computation** for testing
- ✓ Most time-consuming in development cycle
- ✓ Often **hand-craft** and **task dependent** in practice

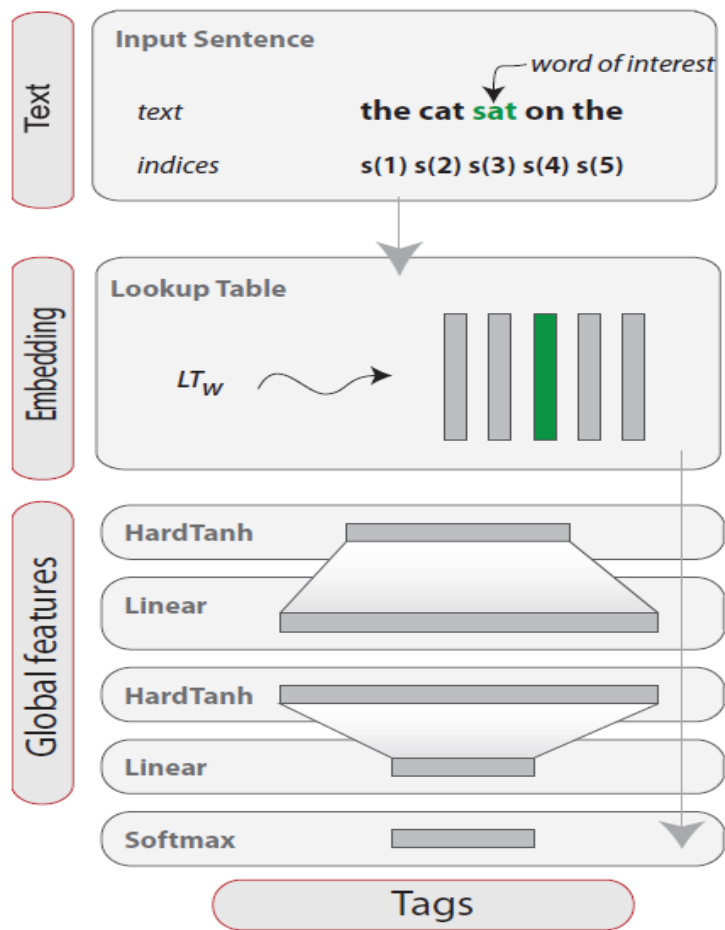


Feature / Representation Learning

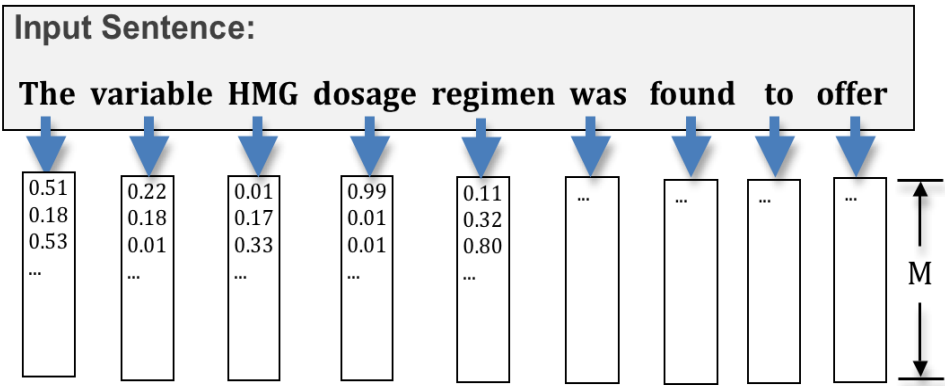
- ✓ Easily **adaptable to new** similar tasks
- ✓ Layerwise representation
- ✓ Layer-by-layer unsupervised training
- ✓ Layer-by-layer supervised training

Deep Learning Based

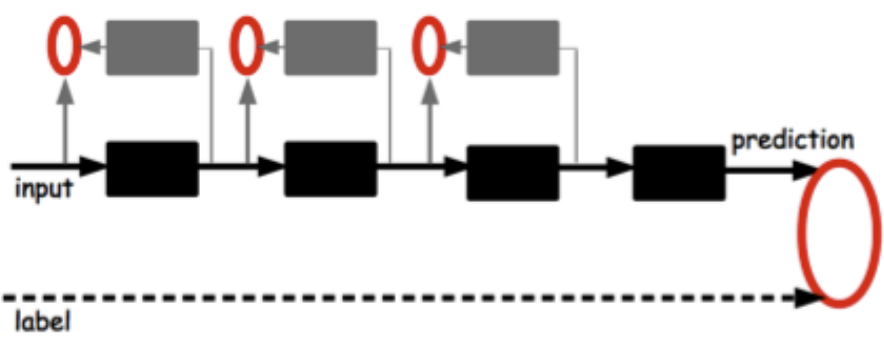
Deep Multi-Layer Learning



Supervised Embedding



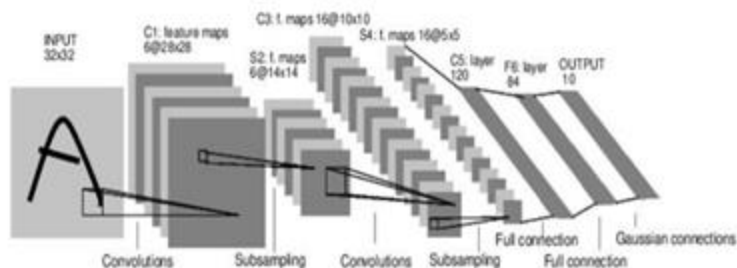
Layer-wise Pretraining



History of ConvNets

1998

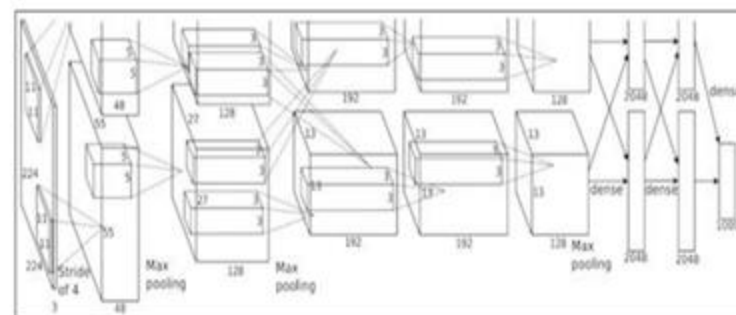
Gradient-based learning applied to document recognition [LeCun, Bottou, Bengio, Haffner]



LeNet-5

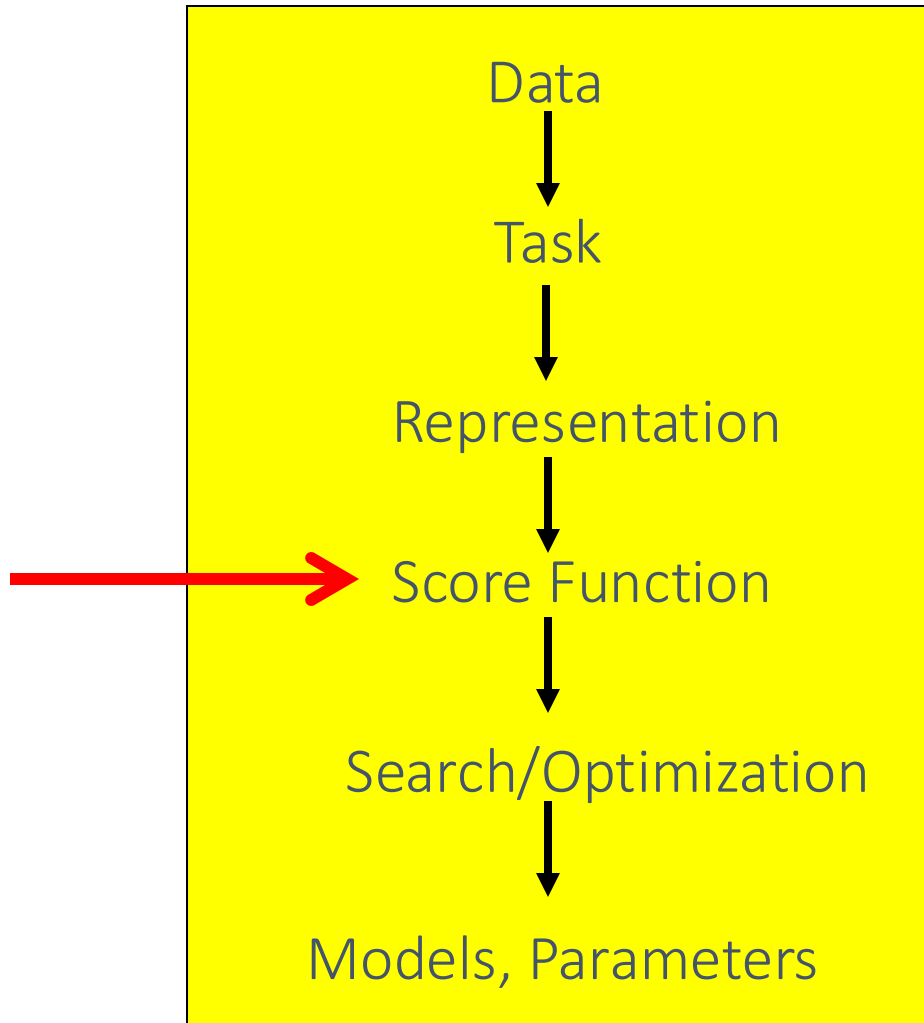
2012

ImageNet Classification with Deep Convolutional Neural Networks [Krizhevsky, Sutskever, Hinton, 2012]



“AlexNet”

Machine Learning in a Nutshell



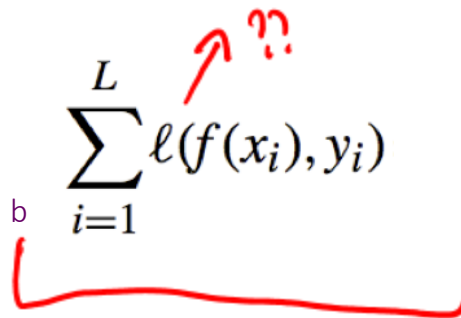
ML grew out of
work in AI

Optimize a
performance criterion
using example data or
past experience,

Aiming to generalize to
unseen data

Basic Concepts

- Training (i.e. learning parameters w, b)
 - Training set includes
 - available examples $x_1, \dots, x_L$
 - available corresponding labels $y_1, \dots, y_L$
 - Find (w, b) by minimizing loss
 - (i.e. difference between y and $f(x)$ on available examples in training set)

$$(W, b) = \underset{W, b}{\operatorname{argmin}} \sum_{i=1}^L \ell(f(x_i), y_i)$$


Basic Concepts

- Loss function

- e.g. hinge loss for binary classification task

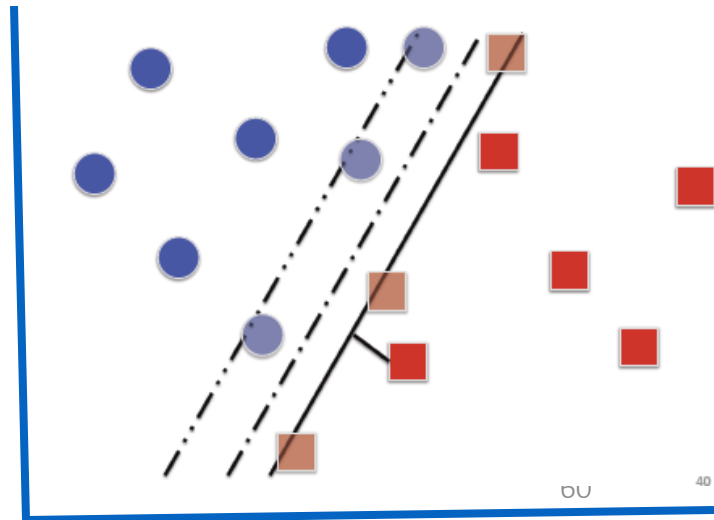
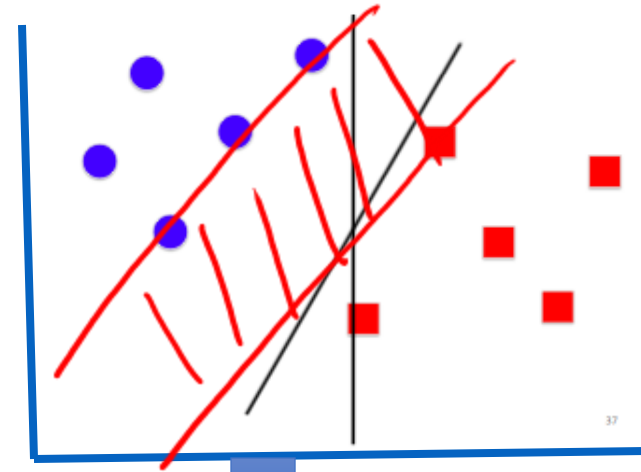
$$\sum_{i=1}^L \ell(f(x_i), y_i) = \sum_{i=1}^L \max(0, 1 - y_i f(x_i)).$$

$|f(x_i) - y_i|$
 $(f(x_i) - y_i)^2$

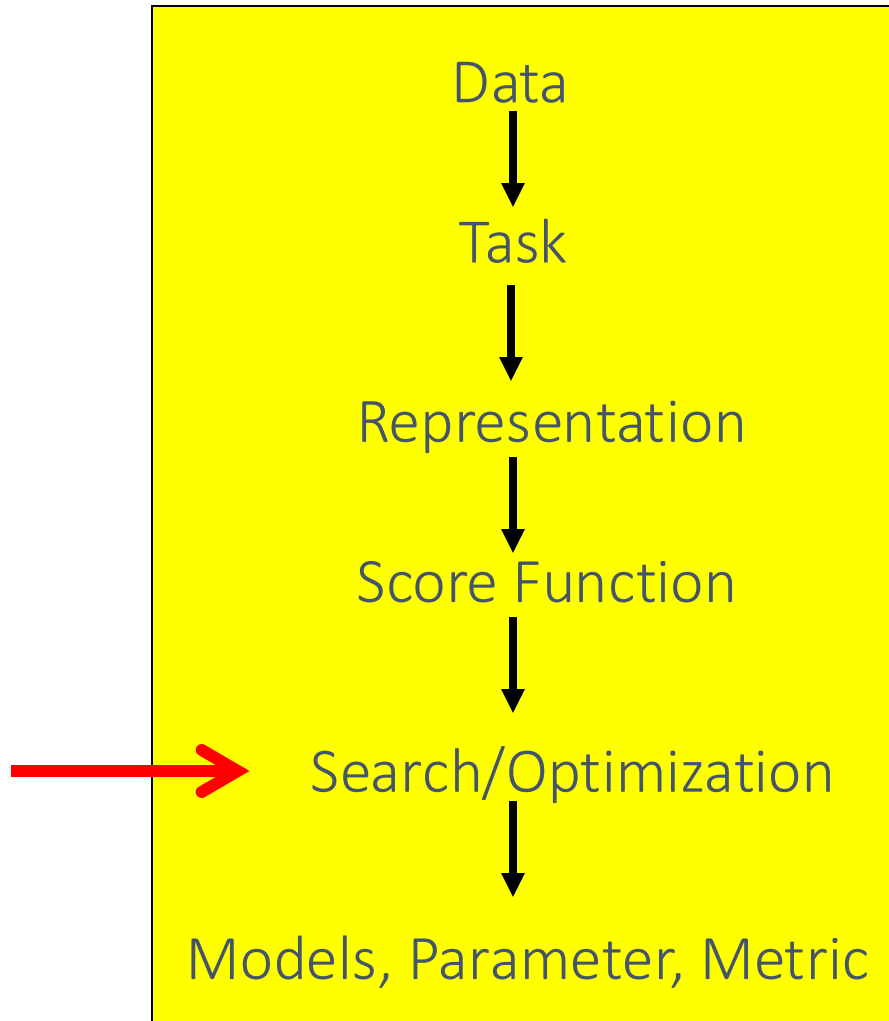
- Regularization

- E.g. additional information added on loss function to control f

$$C \sum_{i=1}^L \ell(f(x_i), y_i) + \frac{1}{2} \|w\|^2,$$



Machine Learning in a Nutshell



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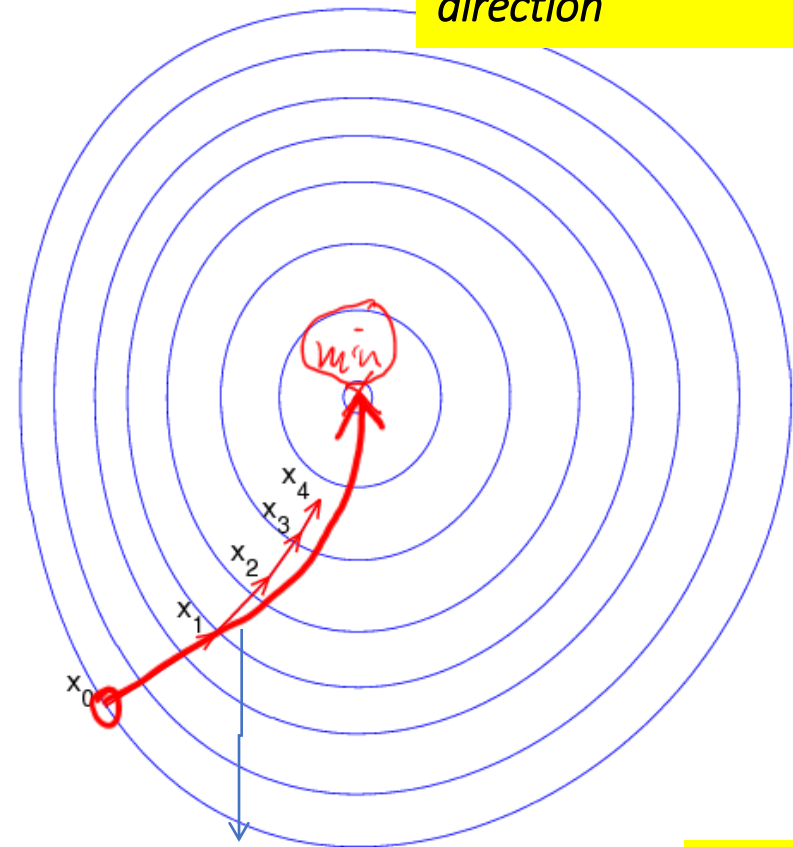
Aiming to generalize to
unseen data

Gradient Descent (Steepest Descent) – contour map view

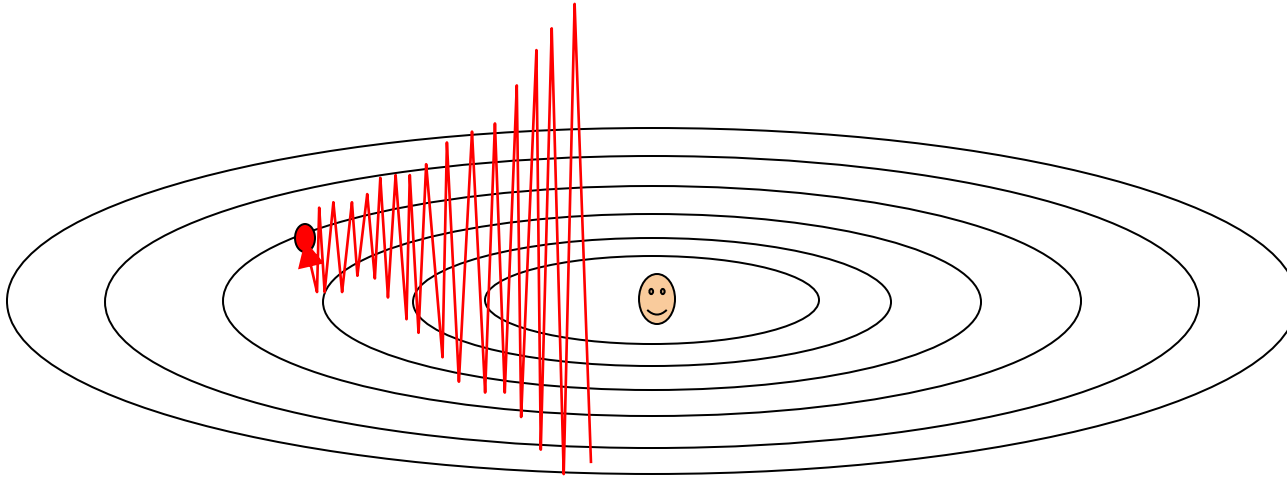
A first-order optimization algorithm.

To find a local minimum of a function using gradient descent, one takes steps proportional to the *negative* of the gradient of the function at the current point.

The gradient (in the variable space) points in the direction of the greatest rate of increase of the function and its magnitude is the slope of the surface graph in that direction



Gradient Magnitudes:



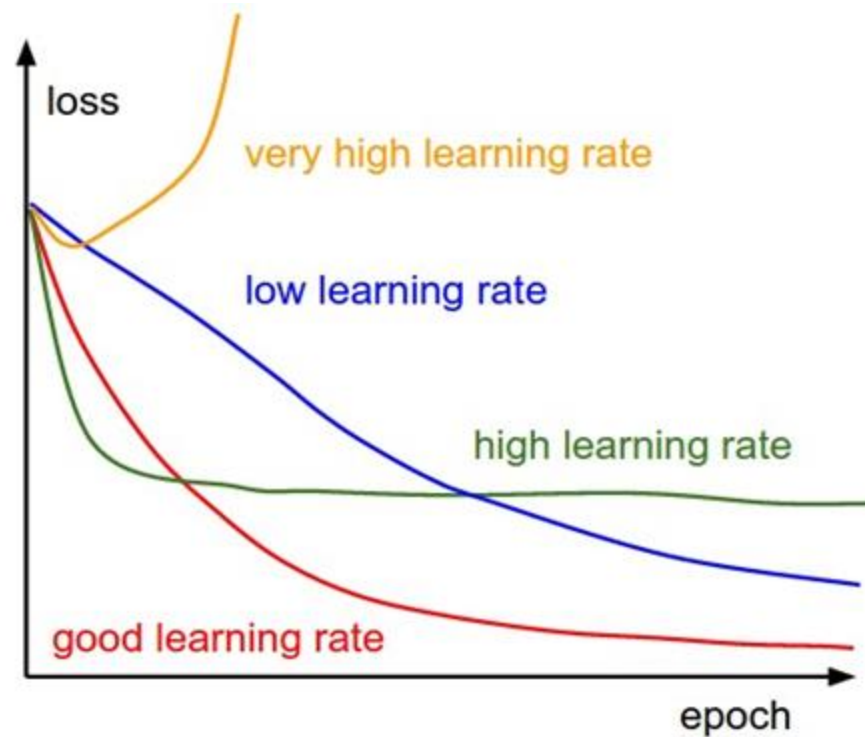
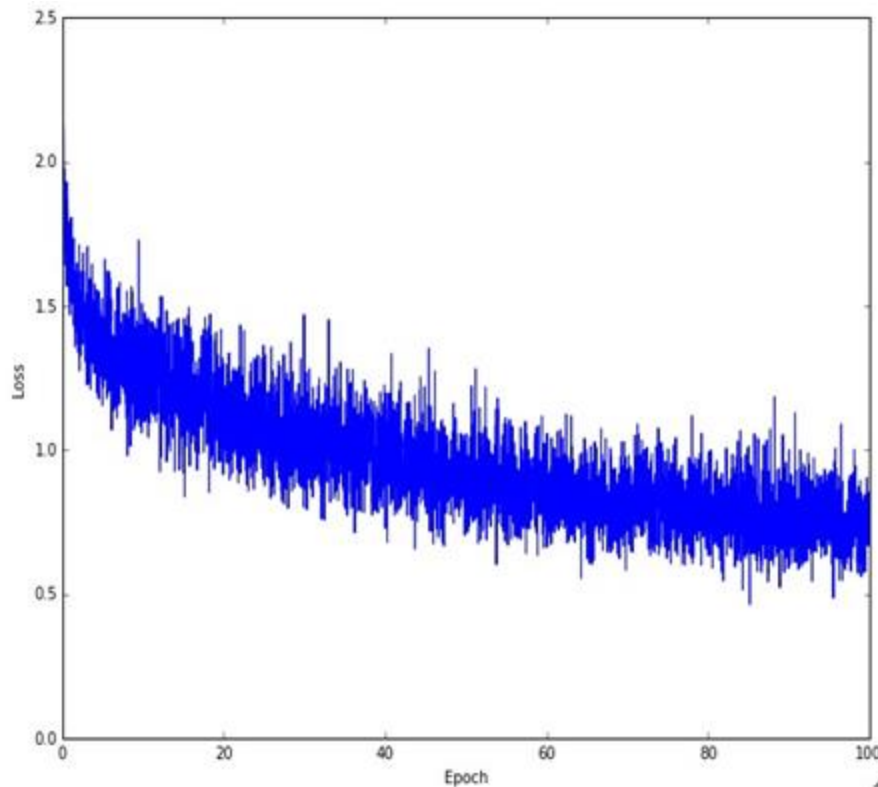
Gradients too big $\rightarrow$ divergence

Gradients too small $\rightarrow$ slow convergence

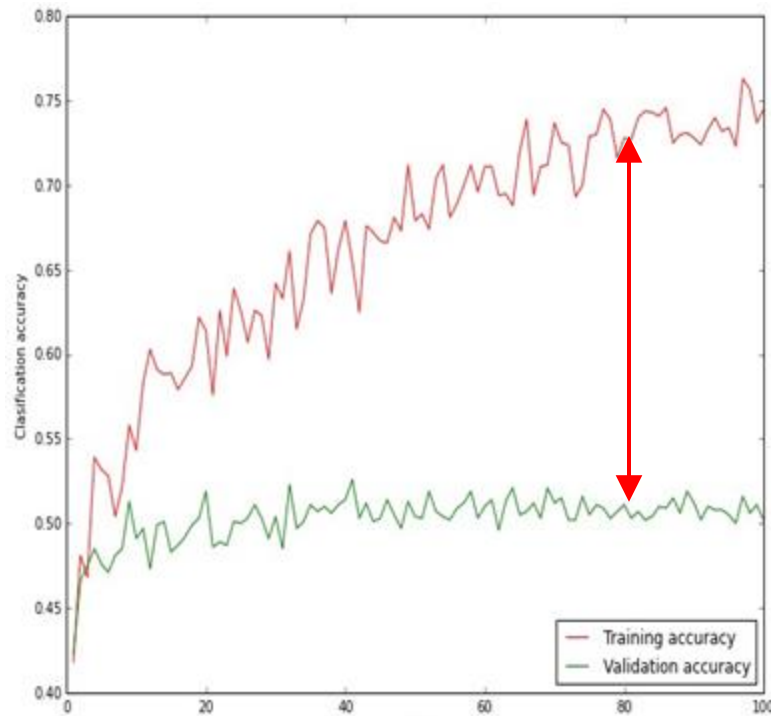
Divergence is much worse!

Many great tools, e.g., Adam
<https://arxiv.org/abs/1609.04747>

Monitor and visualize the loss curve



Monitor and visualize the train / validation loss / accuracy: Bias Variance Tradeoff



big gap = overfitting

=> increase regularization strength?

no gap, e.g. underfitting / both bad

=> increase model capacity?

Large-Scale Machine Learning: **SIZE MATTERS**

LARGE-SCALE

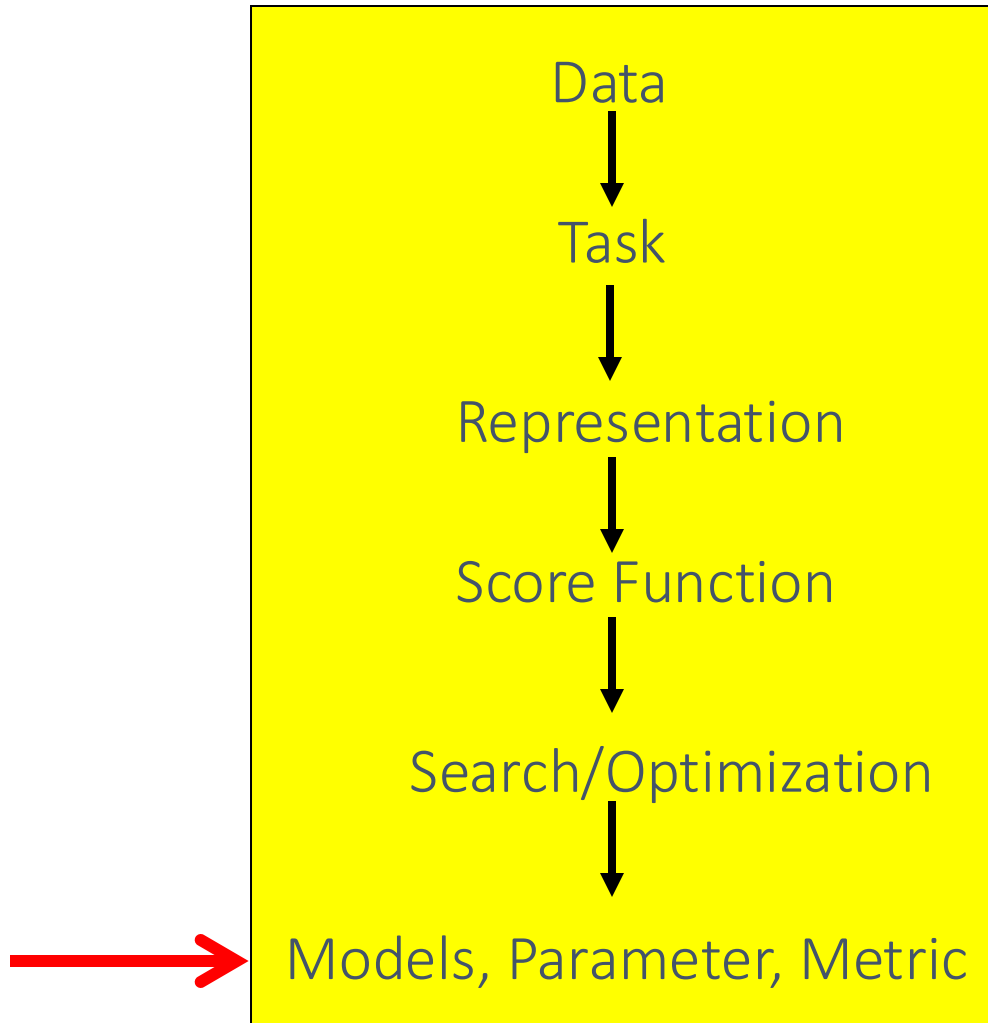


- One thousand data instances
- One million data instances
- One billion data instances
- One trillion data instances

Those are not different numbers,
those **are different mindsets !!!**

Not the focus,
being covered in
my advanced-level
course

Machine Learning in a Nutshell

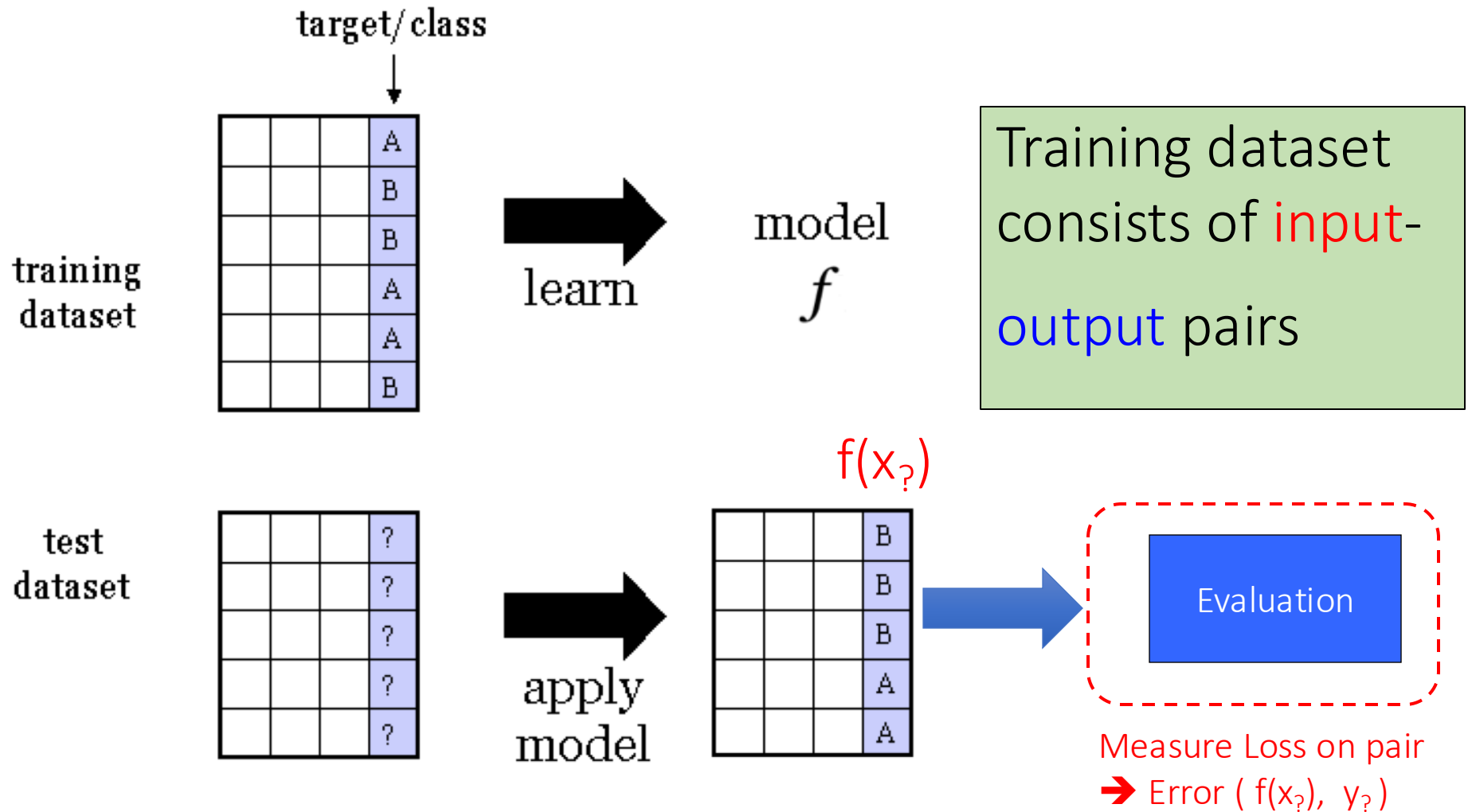


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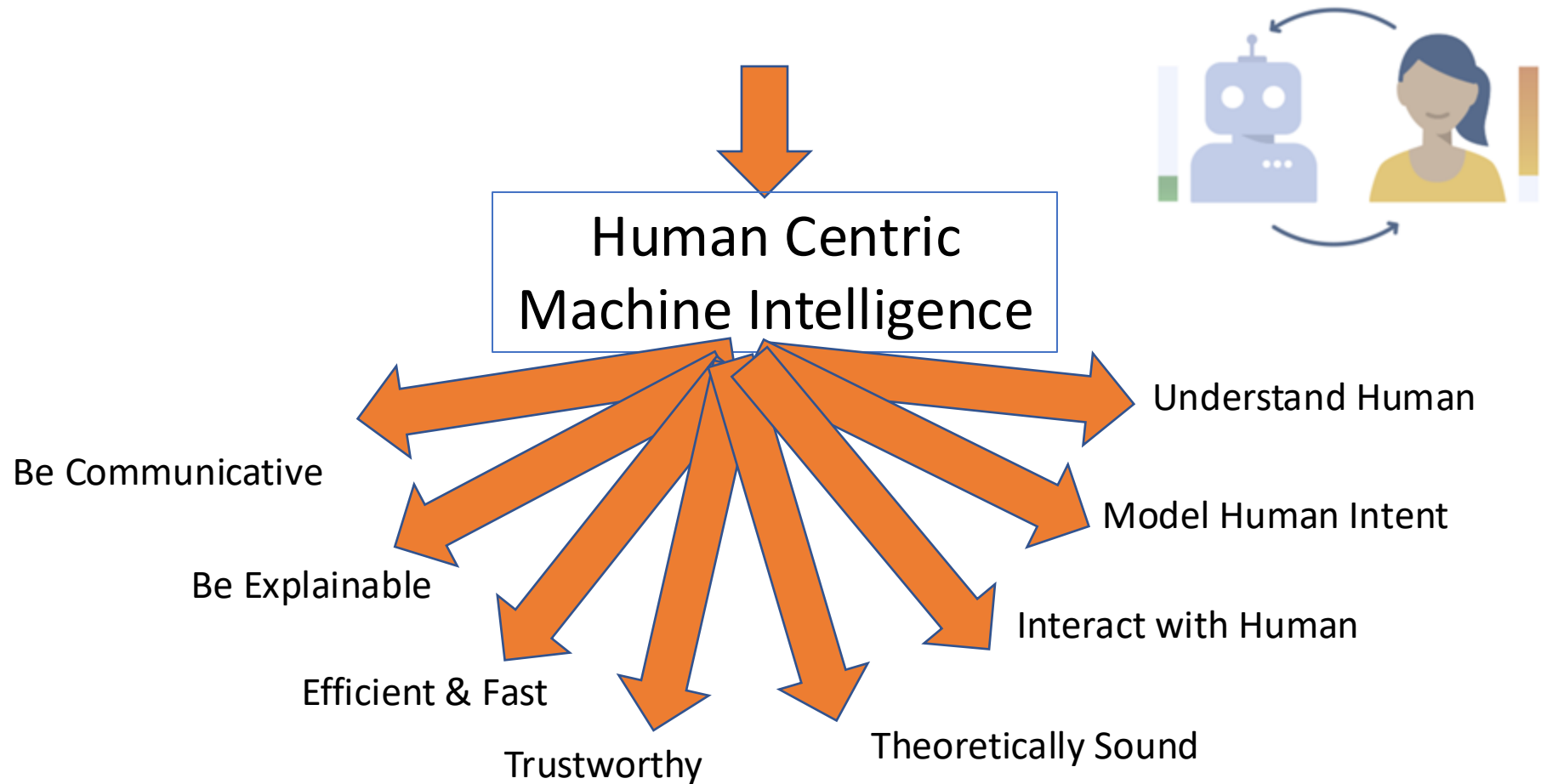
How to know the program works well: Measure Prediction Accuracy on Test Data



Many Metrics for Supervised Classification

Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Hybrid metric useful for unbalanced classes

Beyond Prediction Accuracy: e.g., ML and AI Research @ UVA CS



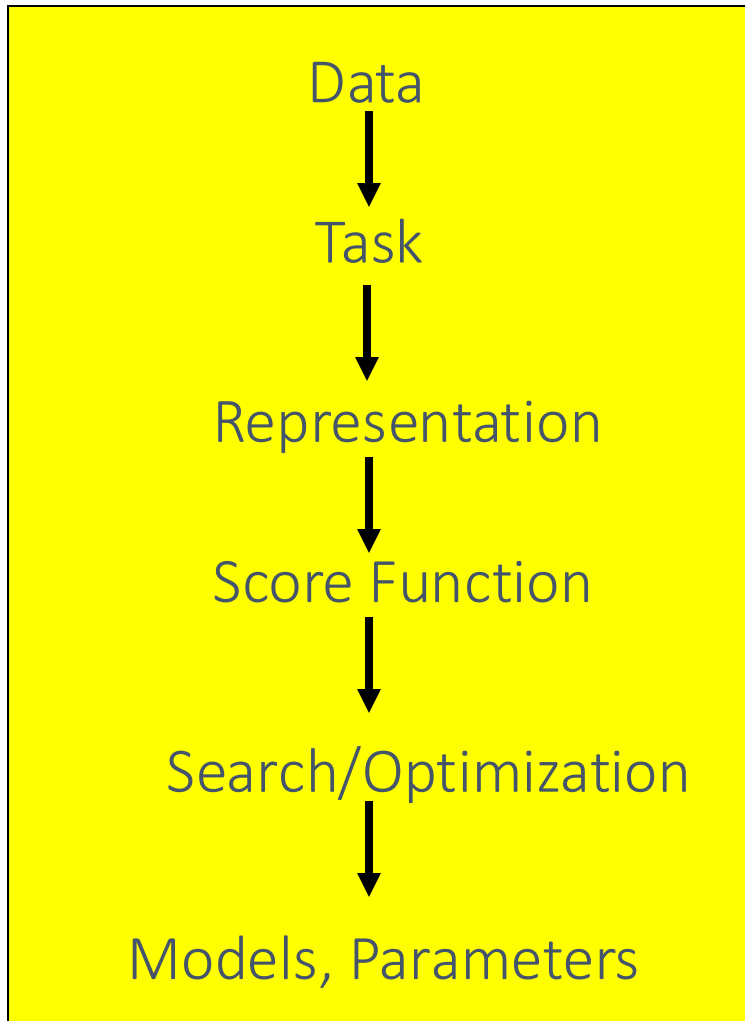
Robustness of DNN, e.g. Adversarial Examples (AE)



$$f(x) \text{ "panda"} + 0.007 \times \underset{\delta}{[noise]} = \text{"gibbon"} \\ f(x+\delta)$$

Example from: Ian J. Goodfellow, Jonathon Shlens, Christian Szegedy. Explaining and Harnessing Adversarial Examples. ICLR 2015.

Machine Learning in a Nutshell



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Rough Sectioning of this Course

- 1. Basic Supervised Regression + Tabular Data
- 2. Basic Deep Learning + 2D Imaging Data
- 3. Advanced Supervised learning + Tabular Data
- 4. Generative and Deep + 1D Sequence Text Data
- 5. Not Supervised

Now open course website Schedule page →

Thank You



References

- Prof. Andrew Moore's tutorials
- Prof. Raymond J. Mooney's slides
- Prof. Alexander Gray's slides
- Prof. Eric Xing's slides
- <http://scikit-learn.org/>
- Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: Springer, 2009.
- Prof. M.A. Papalaskar's slides